

Y A H O

2014.12.11

System engineer

이형석

목차

1. System

2. Payload(Science)

3. ADCS

4. GNC / PPS

5. TTC / CDH

6. EPS

7. TCS / SMS

목차

1. System

1.1 Team 소개

1.2 Objective

1.3 Requirements / Constraints

1.4 Budget

1.5 Subsystem

2. Payload(Science)

3. ADCD

4. GNC / PPS

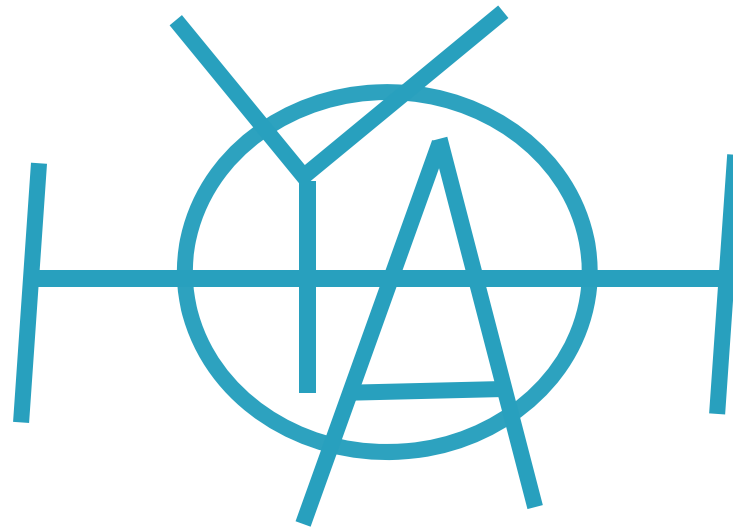
1. TTC / CDH

2. EPS

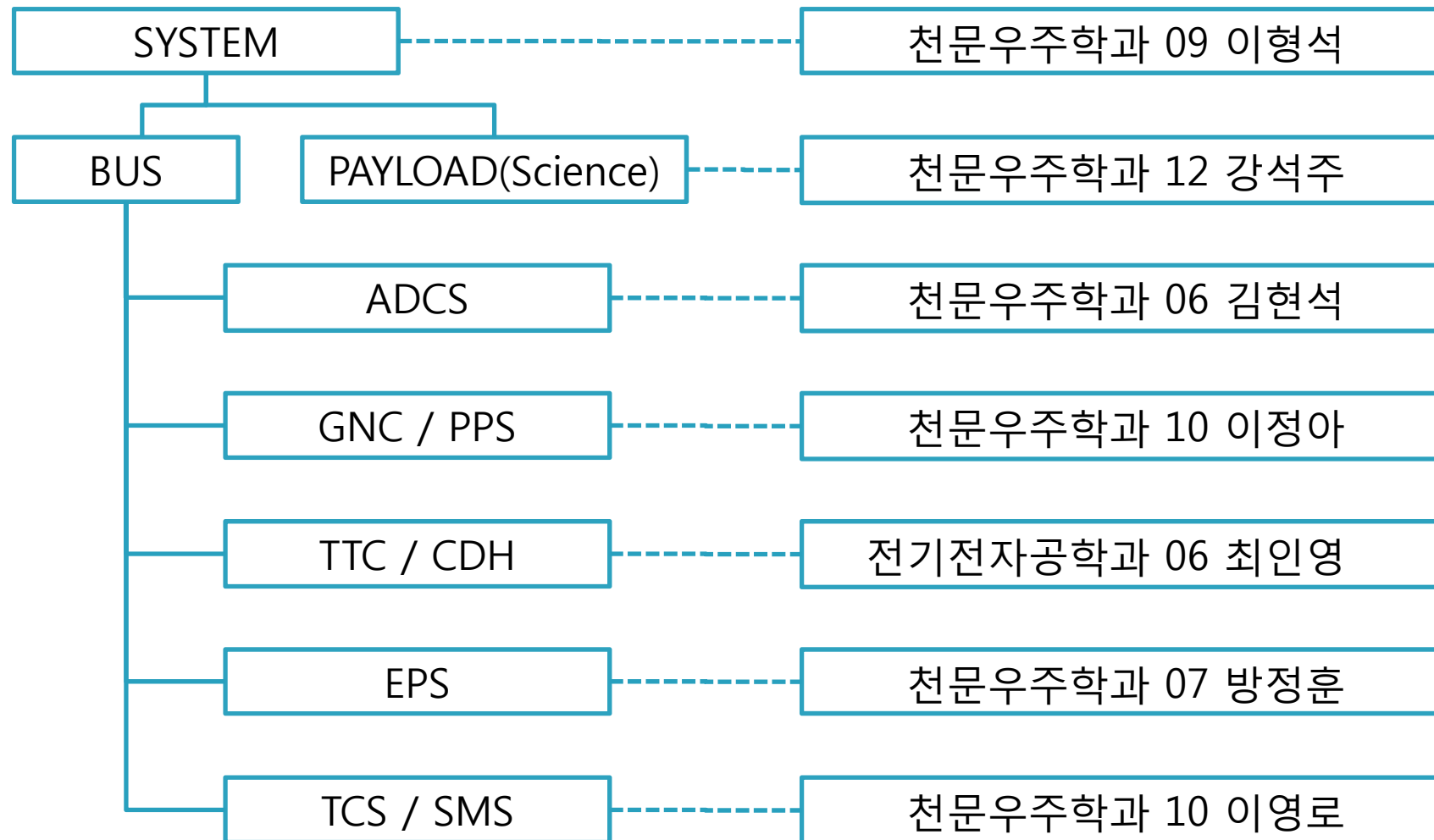
3. TCS / SMS

Y A H O

Yonsei **A**steroid **H**it **O**bservation



Team member



Objective

Primary objective :

소행성(65803 Didymos)의 위성과 임팩터와의 충돌로 인한 **궤도, 모양, 크기, 온도, 지질** 변화를 관측하여 **소행성 충돌 모델**을 기존보다 더 정확하게 만든다.

Secondary objective :

소행성과 위성을 가까이에서 관측하여 그 특징을 **지상에서보다** 더 정확하게 확인한다.
충돌시 발생하는 **먼지와 불출물**의 움직임을 연구한다.

Objective(why 65803 Didymos?)

조건 1. NEAs (Near Earth Asteroids)

조건 2. Binary asteroid

Johnstonsarchive (<http://www.johnstonsarchive.net/astro/asteroidmoons.html>)

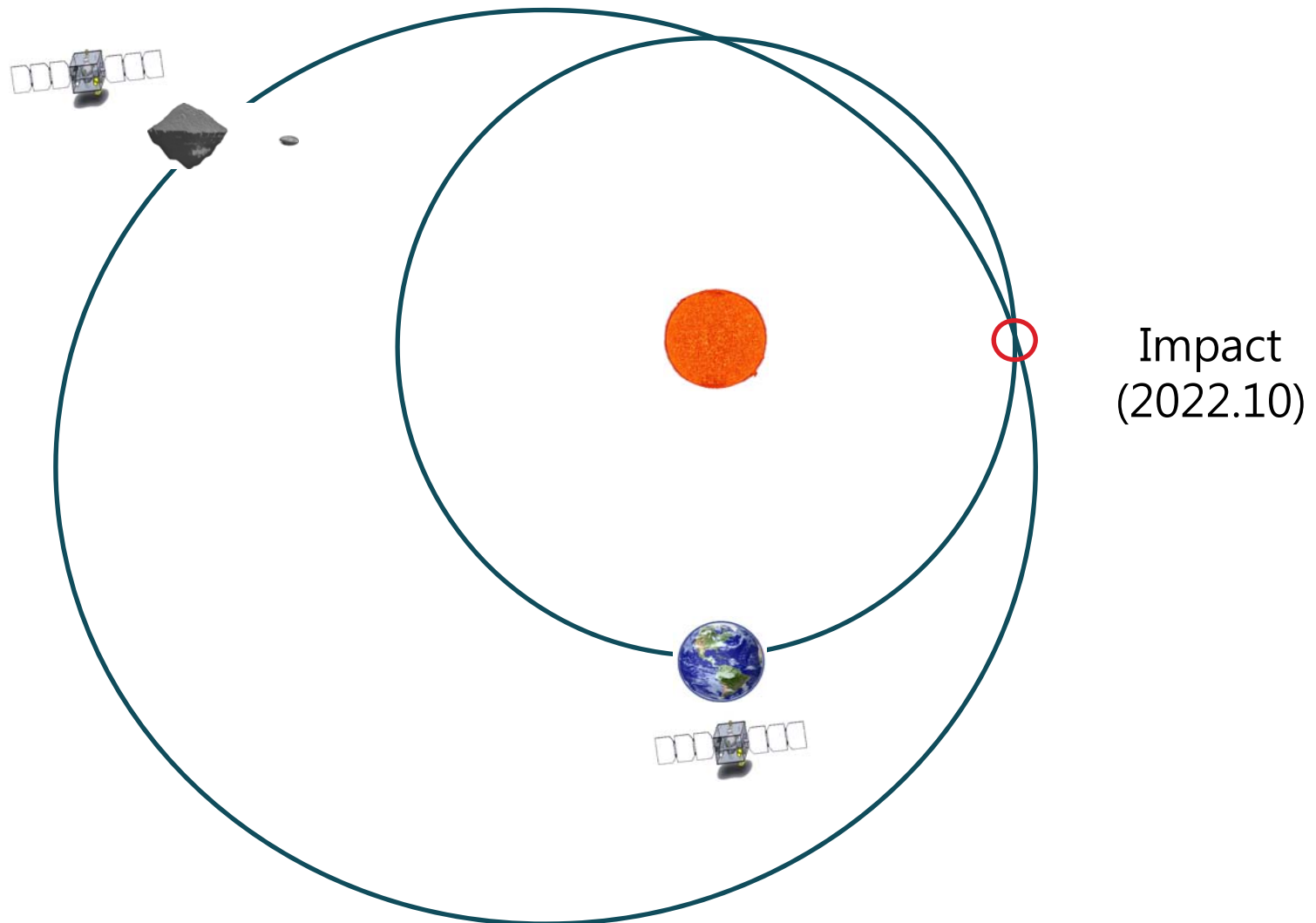
(1862) Apollo	(65803) Didymos	(153591) 2001 SN263	(363027) 1998 ST27	2002 KK8
(1866) Sisyphus	(66063) 1998 RO1	(153958) 2002 AM31	(363067) 2000 CO101	2003 SS84
(3671) Dionysus and S/1997	(66391) 1999 KW4	(162000) 1990 OS	(363599) 2004 FG11	2005 AB
(5143) Heracles	(69230) Hermes	(162483) 2000 PJ5	(374851) 2006 VV2	2006 GY2
(5381) Sekhmet	(85938) 1999 DJ4	(164121) 2003 YT1	(385186) 1994 AW1	2007 DT103
(5646) 1990 TR	(88710) 2001 SL9	(175706) 1996 FG3	(399774) 2005 NB7	2007 LE
(7088) Ishtar	(136617) 1994 CC	(185851) 2000 DP107	1994 CJ1	2008 BT18
(7888) 1993 UC	(136993) 1998 ST49	(276049) 2002 CE26	1994 XD	2008 DG17
(31345) 1998 PG	(137170) 1999 HF1	(285263) 1998 QE2	2000 UG11	2013 WT44
(35107) 1991 VH and S/2008	(138095) 2000 DK79	(311066) 2004 DC	2002 BM26	2014 WZ120

65803 Didymos

Objective(mission scenario)



Objective(mission scenario)



Requirements & Constraints

조건		비고
위성개수	1개	+ 임팩터 1개
크기	높이 7.8m 지름 2.6m	Launch vehicle (VEGA)
무게	> 2500kg	Launch vehicle (VEGA)
온도	200K~300K	관측 환경
통신	2 AU	관측 환경

Requirements & Constraints

일정	2019.8	3년	3개월	2022.10	3개월
특징	Launch	Rendezvous	충돌 전 관측	충돌	충돌 후 관측
궤도			소행성과 15km	소행성과 100Km	소행성과 100Km

Budget

무게

	설계기준	초기설계	최종설계
Payload	20%	80kg	42
ADCS	5%	20kg	36
GNC			
PPS	10%	50kg	
TTC	5%	15kg	7
CDH	5%	5kg	22
EPS	30%	80kg	88
TCS	5%	20kg	31
SMS	20%	80kg	100
총계	100%	350kg	

Budget

전력

	설계기준	초기설계	최종설계
Payload	20%	200W	165
ADCS	20%	200W	120
GNC			
PPS	20%	200W	160
TTC	15%	150W	95
CDH	10%	100W	15
EPS	10%	100W	66(여분)
TCS	10%	100W	100
SMS	5%	50W	8
총계	100%	1000W	729(660)W

SCIENCE & PAYLOAD

2014.12.8

YAHN

강석주

목차

1. SCIENCE

2. PAYLOAD

3. 레이저 고도계

4. 근적외선 분광기

5. 열 적외선 카메라

6. 가시광선 영역 카메라

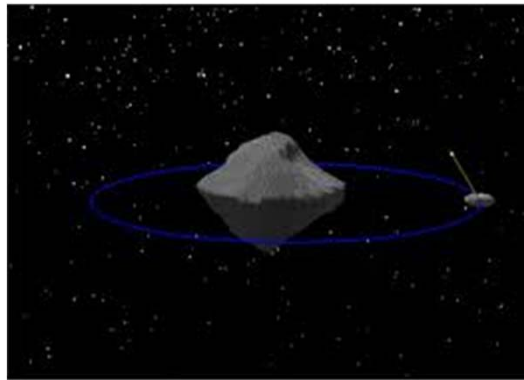
SCIENCE

충돌 과정과 소행성에 대한 연구

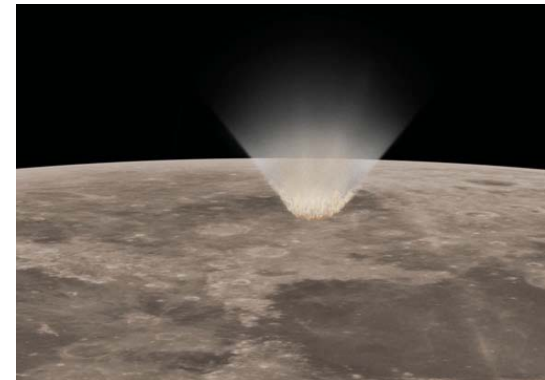
- (1) 충돌 현상 관찰
- (2) Binary System 연구
- (3) Dust and Ejecta Dynamics 연구



Deep impact mission



65803 Didymos

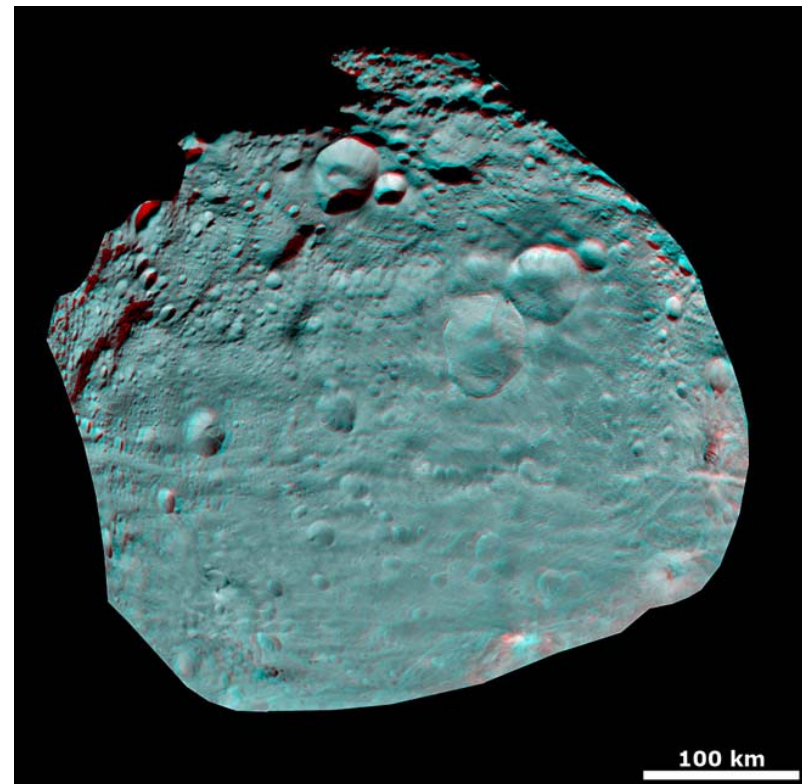


Asteroid ejecta

SCIENCE

관측 가능한 것

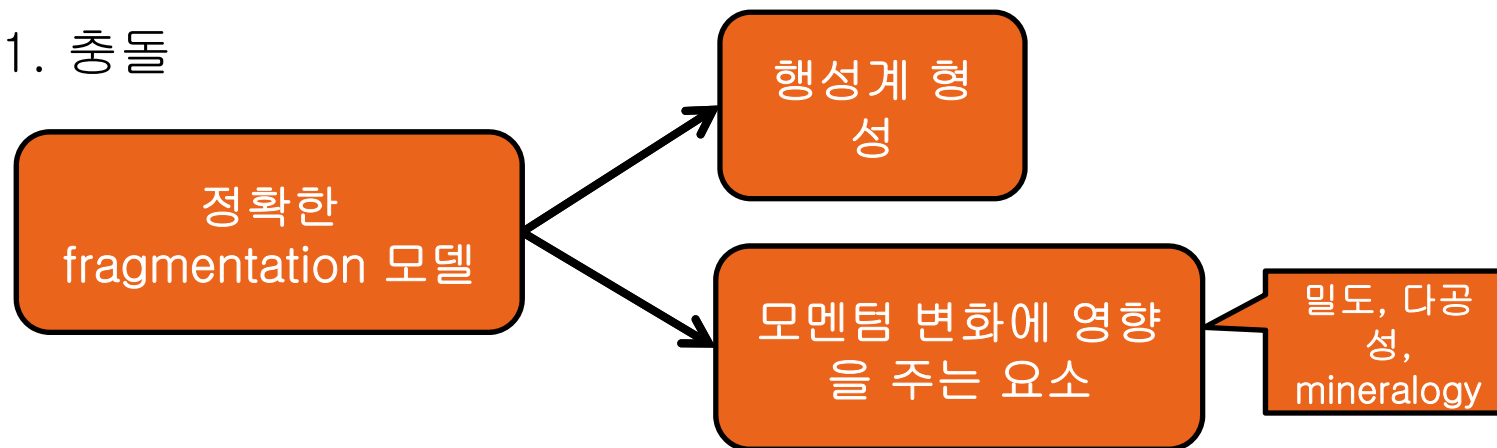
- (1) Fragmentation model
- (2) 크레이터 크기
- (3) 속도 측정
- (4) 소행성 또는 Ejecta의 모양, 크기



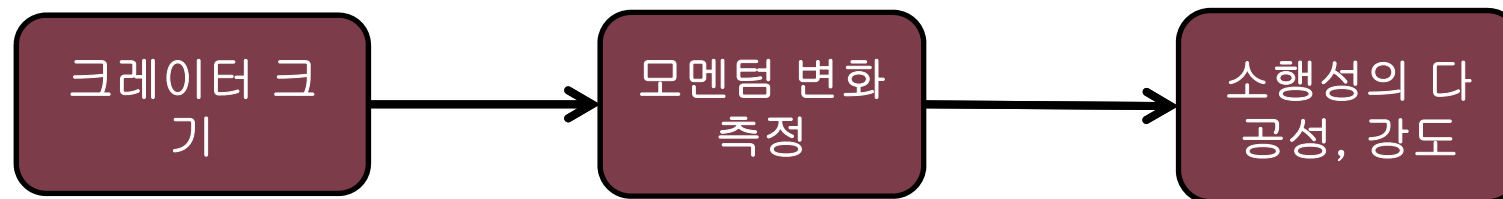
SCIENCE

충돌 현상 관찰

1. 충돌



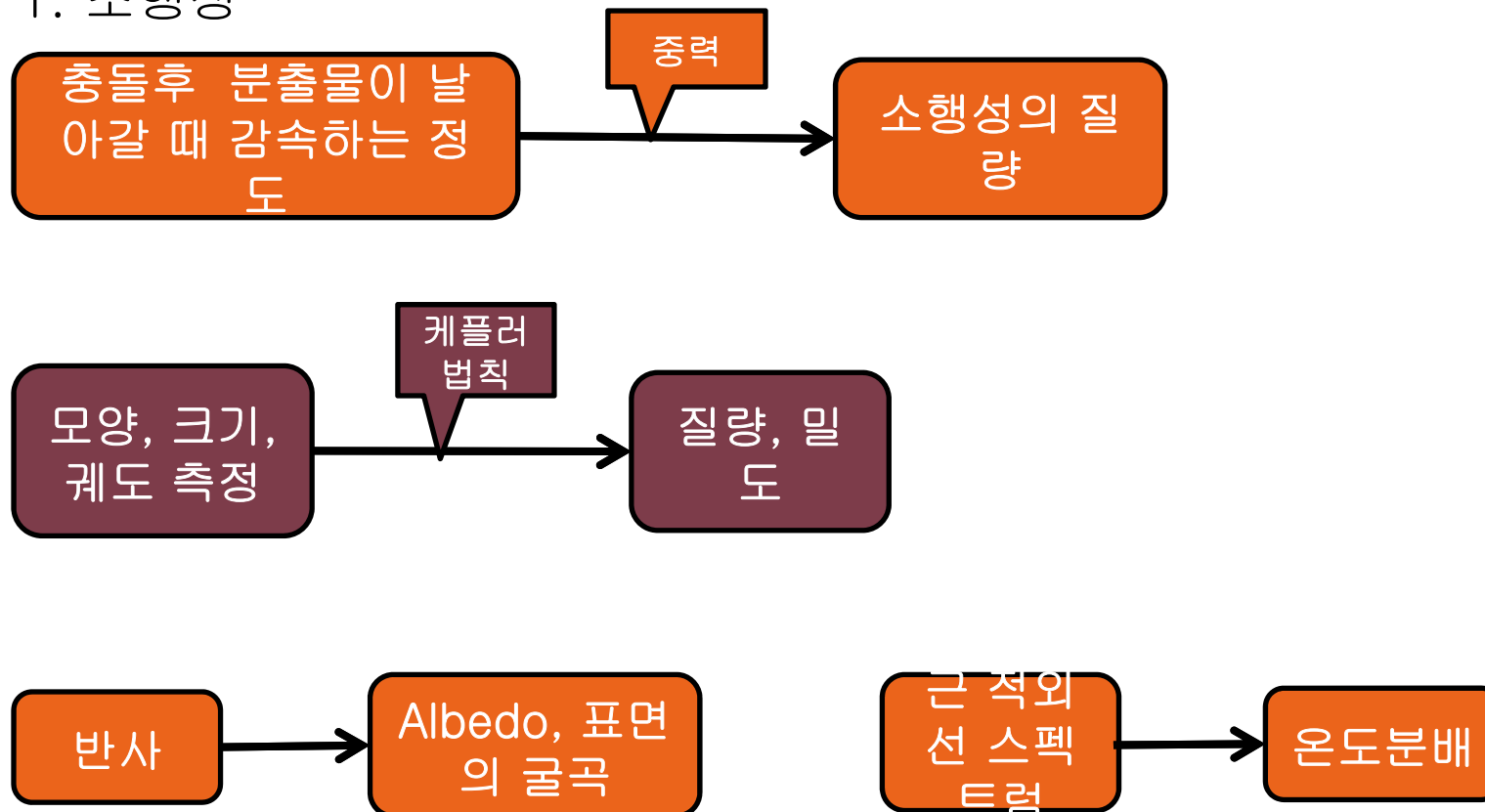
2. 소행성



SCIENCE

Binary System 특성

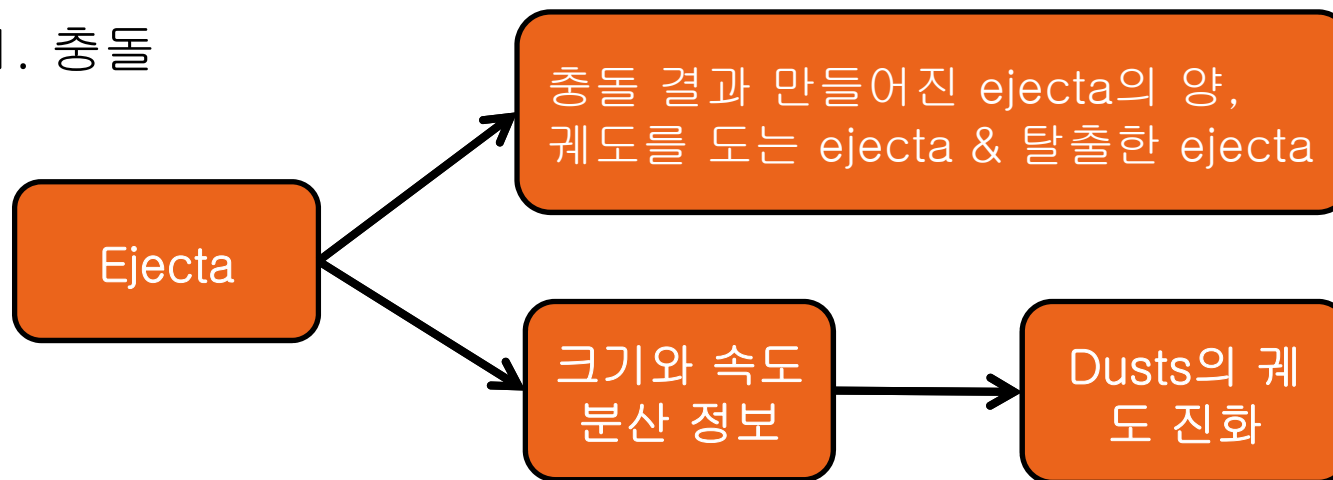
1. 소행성



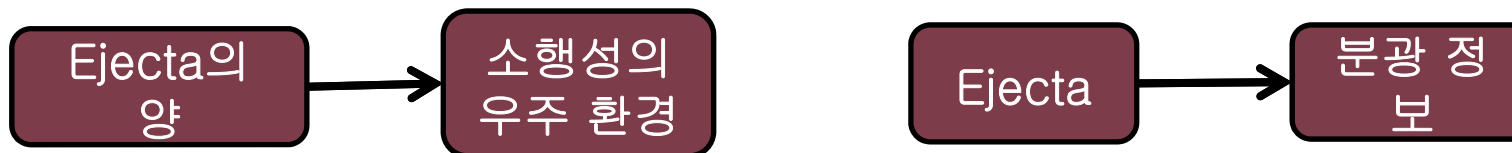
SCIENCE

Dust and Ejecta Dynamics

1. 충돌



2. 소행성



SCIENCE

관측 후 알 수 있는 것

(1) 충돌: 행성계 형성 기원

충돌 후 모멘텀 변화에 영향을 주는 요소

(2) 소행성: 다공성, 강도, 질량, 밀도

albedo, 온도분배

분광 정보

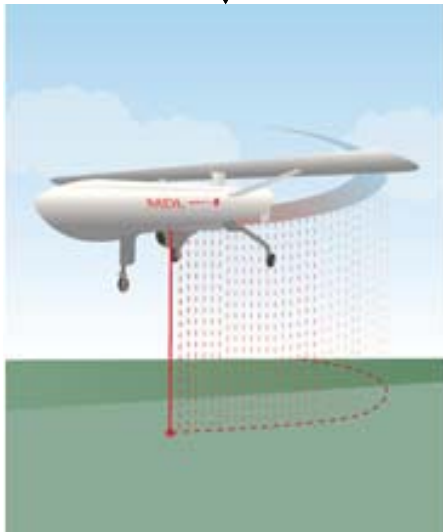
우주 환경



PAYLOAD

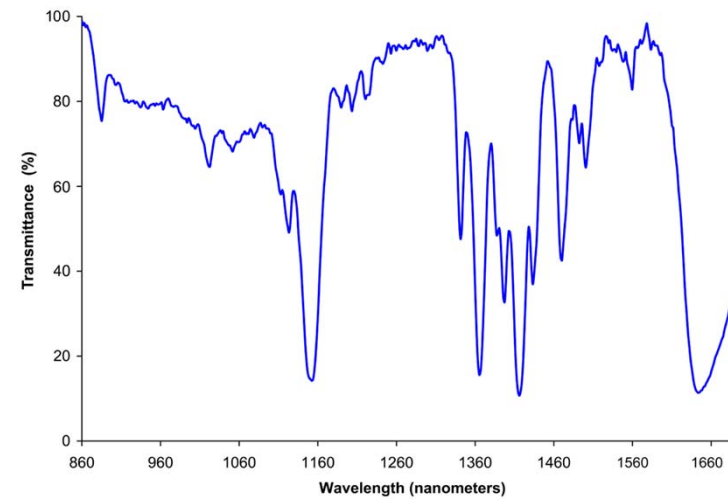
1.레이저 고도계

소행성의 모양,
크레이터의 크기,
속도 측정



2. 근적외선 분광기

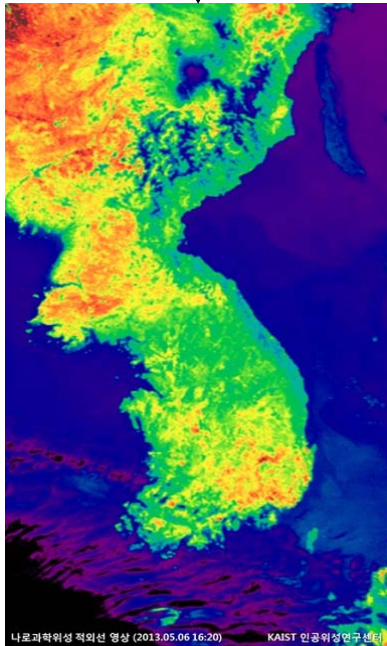
소행성의 화학적
성분



PAYLOAD

3. 열 적외선 카메라

행의 온도와 열,
표면의 거친정도



4. 가시광선 영역 카메라

Albedo, 소행
성의 모양



PAYLOAD

요구조건

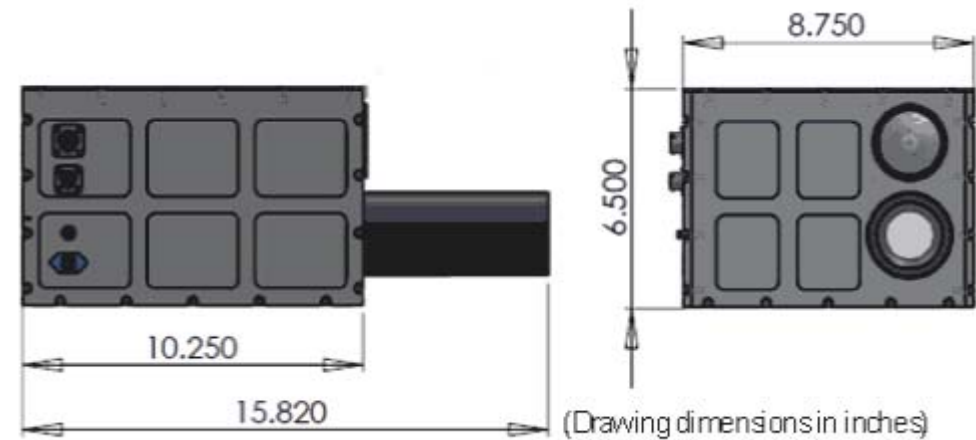
정확도/분해능	레이저고도계	가시광선 영역 카메라
충돌전(15km)	0.01%	20.6265arcsec
충돌후(100km)	0.0015%	3.09397arcsec

분해능	열 적외선 카메라	근 적외선 분광기
충돌 전후	2K	10nm

레이저 고도계

레이저 고도계 성능

정확도	0.0014%
사용시간	10초/10분 충돌 순간부터 20분/1시간
크기	22.2cm*16.5cm*26.5cm
질량	10.24kg
전력	84W
작동온도	-20도 ~ 60도

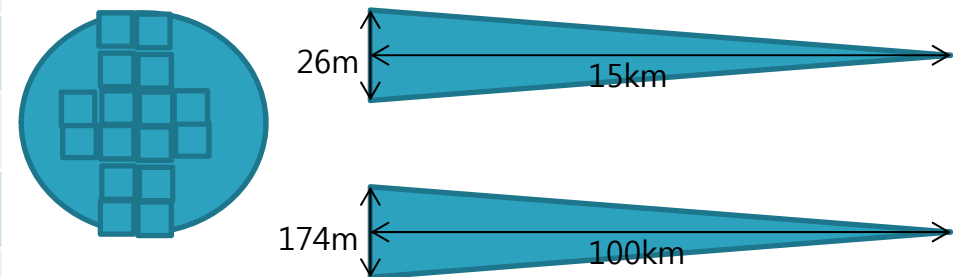
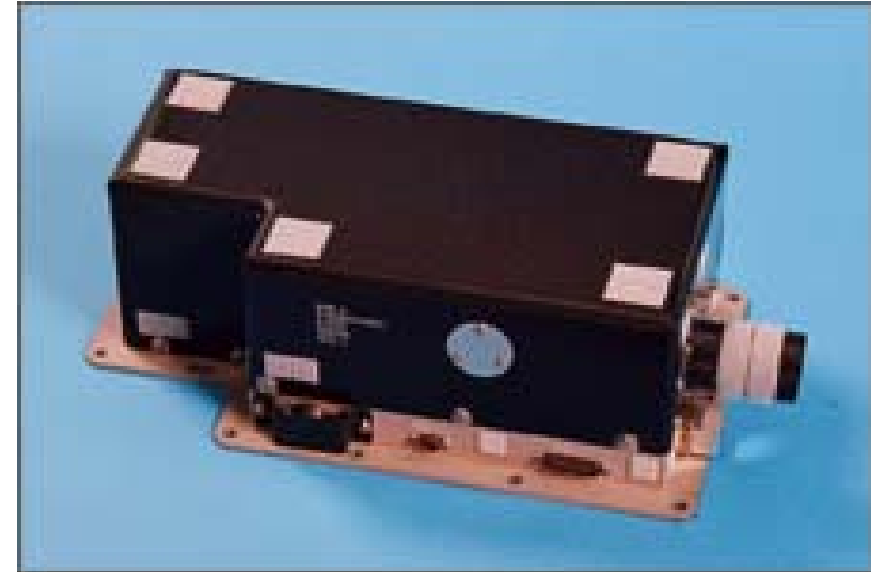


COSMETIC

근적외선 분광기

근적외선 분광기 성능

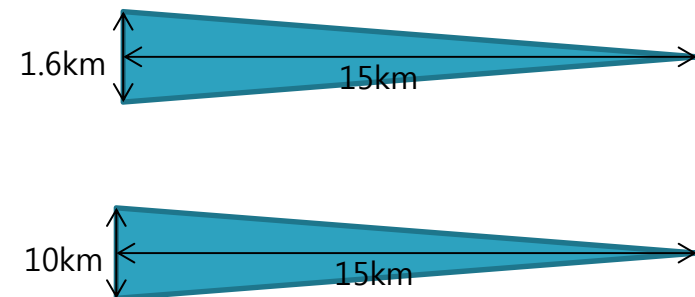
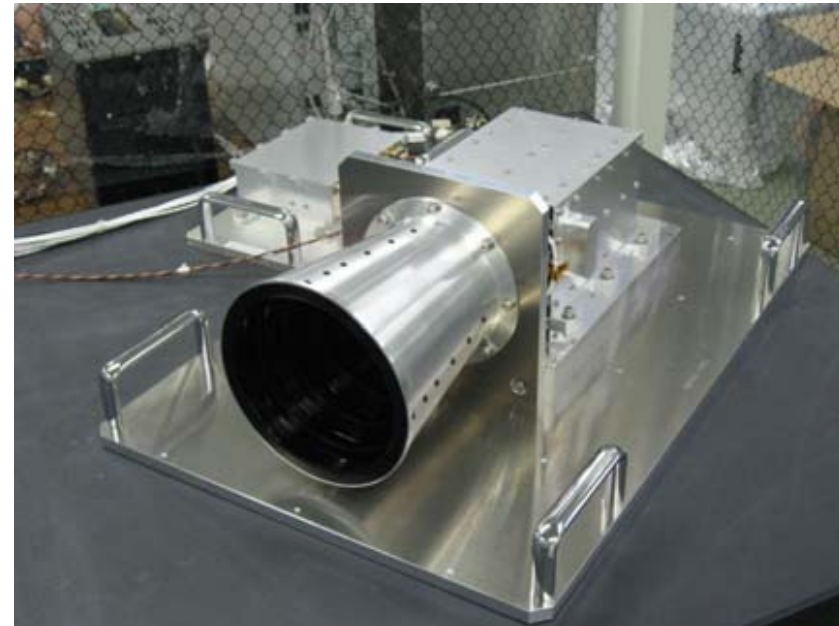
분해능	2~6nm
Spectral range	0.7 μ m~2.1 μ m
FOV	0.1도
사용시간	16장/1일 (15km): 약 2분 노출 충돌 순간부터 하루 동안 연속적으로 분광관측 (100km): 약 100분 노출
크기	32cm*63cm*22cm
질량	15.5kg
전력	15W
작동온도	20도 ~ 110도



열 적외선 카메라

열 적외선 카메라

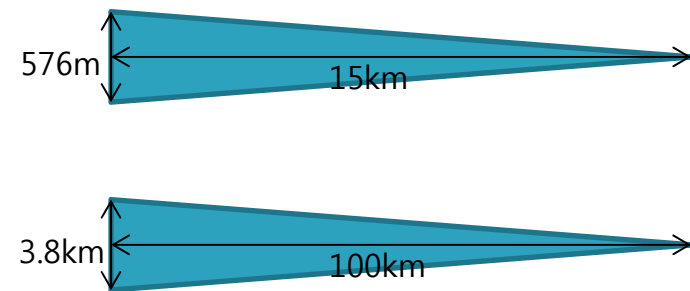
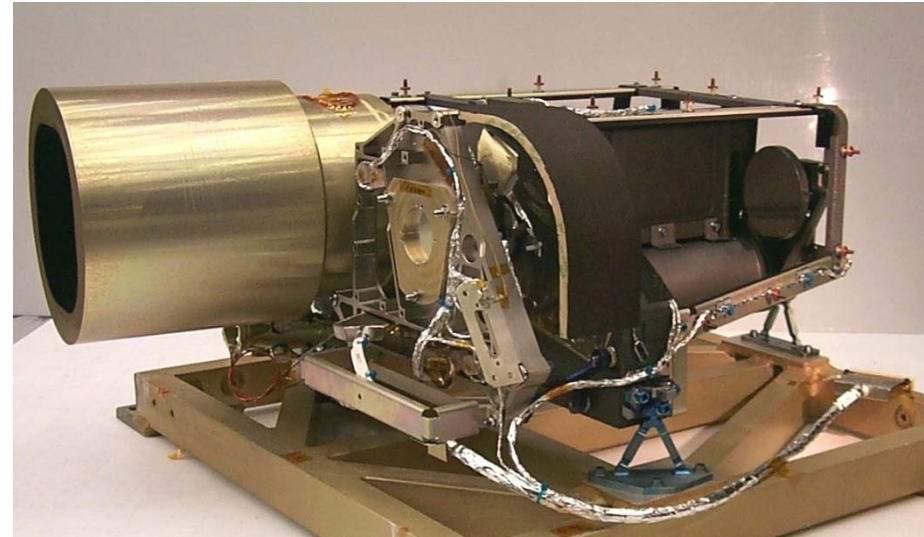
분해능	2K
FOV	8도 * 6도
온도범위	-73도 ~ 127도
사용시간	10장/1일 (15km): 약2분 노출 충돌 순간부터 하루 동안 연속적으로 관측 (100km): 약 100분 노출
크기	16cm*31cm*14cm
질량	3.3kg
전력	22W
작동온도	12도~120도



가시광선 영역 카메라

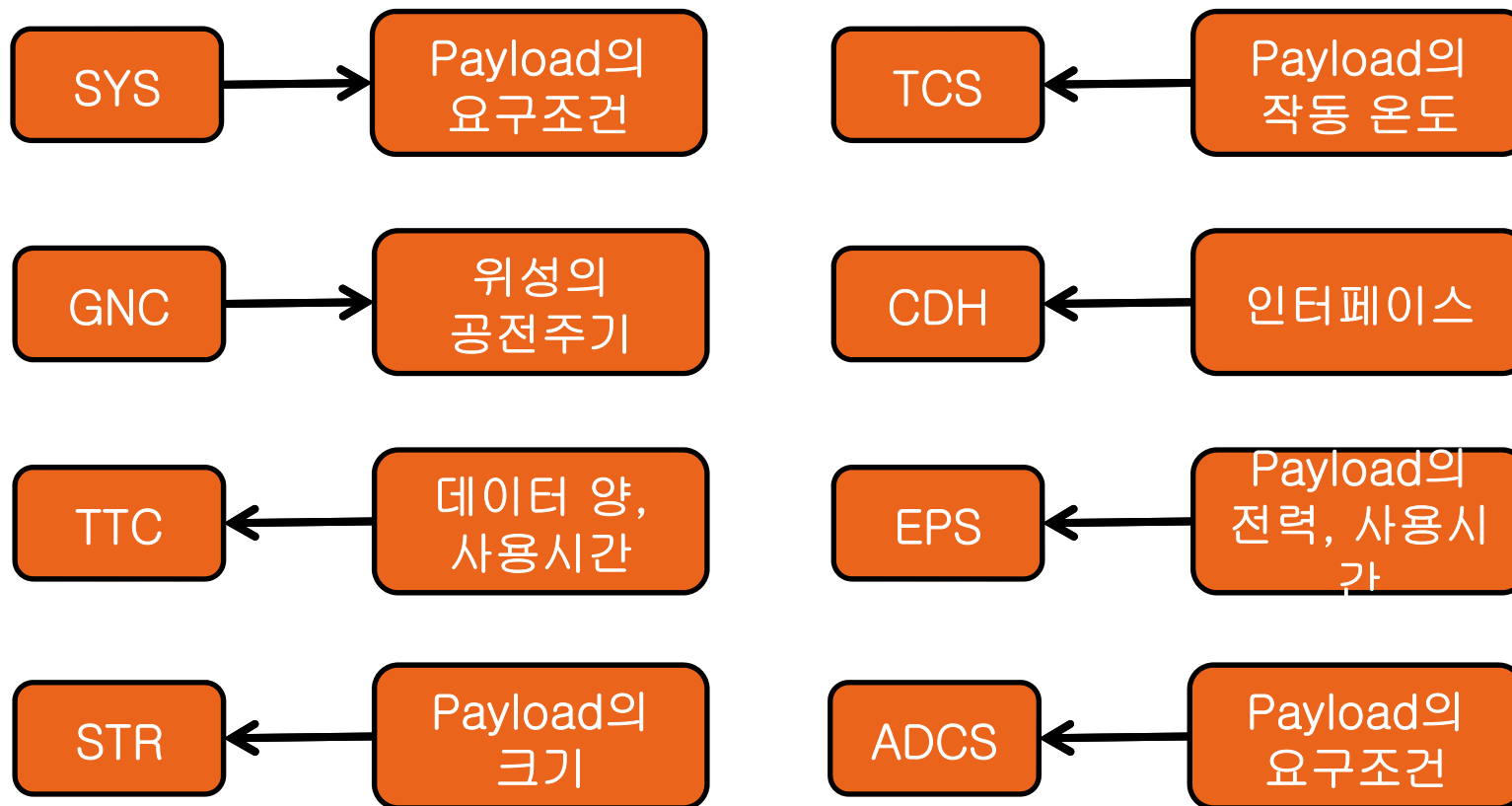
가시광선 영역 카메라

분해능	3.036arcsec
Spectral Range	250nm~1 μ m
FOV	2.2도
사용시간	10장/1일 (15km): 7초 노출 충돌 순간부터 100장 (100km): 약 5분 노출
크기	34cm*43cm*22cm
질량	13.2kg
전력	44W
작동온도	-113도~27도



PAYLOAD

서브시스템간의 상호작용



SCIENCE & PAYLOAD

참고문헌

- http://www.cailidar.com/products/laser_altimeter.html (레이저 고도계)
- http://pdssbn.astro.umd.edu/holdings/ro-a-osiwac-3-ast1-steinsflyby-v1.4/document/osiris_ssr/osiris_ssr.pdf (가시광선영역 카메라)
- <http://www.lpi.usra.edu/meetings/lpsc2012/pdf/1498.pdf> (적외선 카메라)
- <http://arxiv.org/ftp/arxiv/papers/0912/0912.4797.pdf> (노출시간 정보)
- http://global.jaxa.jp/projects/sat/muses_c/topics.html (노출시간 정보)
- AIDA_MissionRationale_InterimRelease (Science)

감사합니다

ADCS

2014. 12. 11

김현석

ADCS Control Mode

- **Sign-up Mode**
- **Normal Mode**
- **Momentum Unlading Mode**
- **Mission Mode**
- **Safe Mode**

ADCS 요구조건

- **Mission (Science/Payload)**
 - Accuracy < 3 arcsec
- **Communications (TTC/C&DH)**
 - Antenna Pointing Accuracy < 0.25 deg
- **Propulsion**
 - Thruster Power > 1N
- **Structure**
 - Basic Information of structure size
 - Mass < 50kg(total)
- **EPS**
 - Power < 100W(total)

ADCS Control Mode

- **Sign-up Mode**
 - **Attitude Determination 요구조건**
 - Accuracy 0.5 deg, 1 σ , real-time
 - Range 10 deg within each Axis
 - **Attitude Control 요구조건**
 - Accuracy 0.5 deg, 1 σ , real-time
 - Settling Time 1h
 - Slew Rate <0.1deg / 1min
 - **Sensor & Actuator**
 - CSS, Magnetometer, Gyro, Wheel, Thruster

ADCS Control Mode

- Normal Mode
 - Attitude Determination 요구조건
 - Accuracy 0.1 deg, 1 σ , real-time
 - Range -Z Axis - 0.1deg / GS
 - Attitude Control 요구조건
 - Accuracy 0.1 deg, 1 σ , real-time
 - Sensor & Actuator
 - Magnetometer, Gyro, Wheel

ADCS Control Mode

- **Momentum Unlading Mode**
 - **Attitude Control 요구조건**
 - Accuracy 10 deg, 기존자세
 - Settling Time 1시간 이내
 - **Sensor & Actuator**
 - Gyro, Wheel, Thruster

ADCS Control Mode

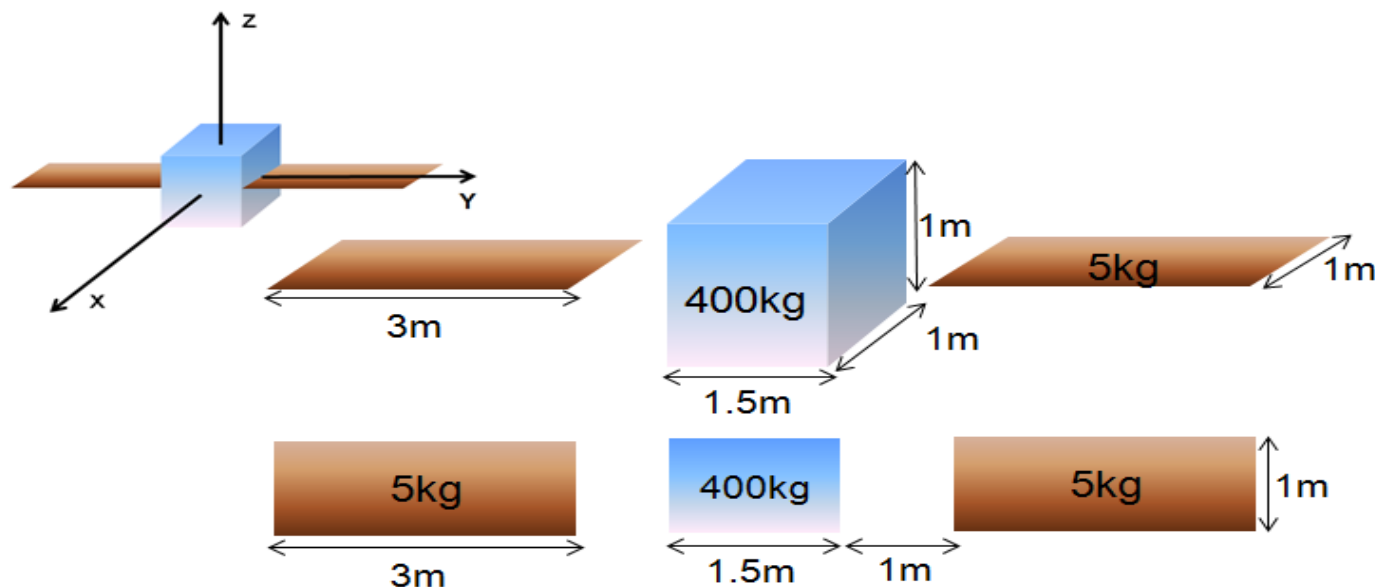
- **Mission Mode**
 - **Attitude Determination 요구조건**
 - Accuracy 3.09397arcsec, 100km / 1.5m
 - Range -X Axis 3arcsec, Target
 - **Attitude Control 요구조건**
 - Accuracy 3arcsec
 - Slew Rate <0.05deg / 1s
 - **Sensor & Actuator**
 - CSS, Startracker, Gyro, Wheel

ADCS Control Mode

- **Safe Mode**
 - **Attitude Determination 요구조건**
 - Accuracy 0.1 deg, 1 σ , real-time
 - Range -Z Axis 0.1deg, GS
 - **Attitude Control 요구조건**
 - Accuracy 0.1 deg, 1 σ , real-time
 - **Sensor & Actuator**
 - CSS, Magnetometer, Gyro, Wheel

Disturbance Environment

- Moment of Inertia



– 가정

- 위성의 내부는 균일한 밀도로 이루어져 있다.
- 위성의 무게중심은 기하학적인 무게중심과 같다.

Disturbance Environment

- Moment of Inertia

$$I_X = \frac{1}{12} \times M_{\text{cube}} \times (a^2 + a^2) + 2 \times \left\{ \left(\frac{1}{12} \times M_{\text{panel}} \times b^2 \right) + (M_{\text{panel}} \times c^2) \right\}$$

$$I_X = 150.6250 \text{ kg} \cdot \text{m}^2$$

$$I_Y = \frac{1}{12} \times M_{\text{cube}} \times (a^2 + a^2) + 2 \times \left(\frac{1}{12} \times M_{\text{panel}} \times b^2 \right)$$

$$I_Y = 38.3333 \text{ kg} \cdot \text{m}^2$$

$$I_Z = \frac{1}{12} \times M_{\text{cube}} \times (a^2 + a^2) + 2 \times \left\{ \left(\frac{1}{12} \times M_{\text{panel}} \times (b^2 + c^2) \right) + (M_{\text{panel}} \times d^2) \right\}$$

$$I_Z = 151.1458 \text{ kg} \cdot \text{m}^2$$

Disturbance Environment

- Moment of Inertia

$$I_X = \frac{1}{12} \times M_{\text{cube}} \times (a^2 + a^2) + 2 \times \left\{ \left(\frac{1}{12} \times M_{\text{panel}} \times b^2 \right) + (M_{\text{panel}} \times c^2) \right\}$$

$$I_X = 150.6250 \text{ kg} \cdot \text{m}^2$$

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$$I_Y = \frac{1}{12} \times M_{\text{cube}} \times (a^2 + a^2) + 2 \times \left\{ \left(\frac{1}{12} \times M_{\text{panel}} \times (b^2 + c^2) \right) + (M_{\text{panel}} \times d^2) \right\}$$

$$\mathbf{I_Y = 38.3333 \text{ kg} \cdot \text{m}^2}$$

$$I_Y = 38.3333 \text{ kg} \cdot \text{m}^2$$

$$\mathbf{I_Z = 151.1458 \text{ kg} \cdot \text{m}^2}$$

$$I_Z = \frac{1}{12} \times M_{\text{cube}} \times (a^2 + a^2) + 2 \times \left\{ \left(\frac{1}{12} \times M_{\text{panel}} \times (b^2 + c^2) \right) + (M_{\text{panel}} \times d^2) \right\}$$

$$I_Z = 151.1458 \text{ kg} \cdot \text{m}^2$$

Disturbance Environment

Gravity Gradient

$$\mathbf{T}_g = 1.9022 \times 10^{-4} \mathbf{N} \cdot \mathbf{m}$$

Solar Radiation

$$\mathbf{T}_{sp} = 5.1262 \times 10^{-5} \mathbf{N} \cdot \mathbf{m}$$

Magnetic Field Disturbance

$$\mathbf{T}_m = 6.4850 \times 10^{-4} \mathbf{N} \cdot \mathbf{m}$$

Aerodynamic Disturbance

$$\mathbf{T}_a = 4.5000 \times 10^{-5} \mathbf{N} \cdot \mathbf{m}$$

Disturbance Environment

$$T_g = 1.9022 \times 10^{-4} \text{N} \cdot \text{m}$$

$$T_{sp} = 5.1262 \times 10^{-5} \text{N} \cdot \text{m}$$

$$T_m = 6.4850 \times 10^{-4} \text{N} \cdot \text{m}$$

$$T_a = 4.5000 \times 10^{-5} \text{N} \cdot \text{m}$$

$$T_{\text{disturbance}} = T_g + T_{sp} + T_m + T_a$$

$$T_{\text{disturbance}} = 1.9022 \times 10^{-4} + 5.1262 \times 10^{-5} + 6.4850 \times 10^{-4} + 4.5000 \times 10^{-5}$$

Disturbance Environment

$$T_g = 1.9022 \times 10^{-4} \text{ N} \cdot \text{m}$$

$$T_{sp} = 5.1262 \times 10^{-5} \text{ N} \cdot \text{m}$$

$$T_m = \mathbf{T_{disturbance} = 9.3498 \times 10^{-4} \cong 1.0000 \times 10^{-3} \text{ N} \cdot \text{m}}$$

$$T_a = 4.5000 \times 10^{-5} \text{ N} \cdot \text{m}$$

$$T_{\text{disturbance}} = T_g + T_{sp} + T_m + T_a$$

$$T_{\text{disturbance}} = 1.9022 \times 10^{-4} + 5.1262 \times 10^{-5} + 6.4850 \times 10^{-4} + 4.5000 \times 10^{-5}$$

Sizing Reaction Wheel

- Slew Torque for Reaction Wheel
 - 1deg /20sec를 위한 최소 토크

$$\frac{\theta}{2} = \frac{1}{2} \frac{T}{I} \left(\frac{t}{2} \right)^2$$

$$I_z (= 151.1458 \text{ kg} \cdot \text{m}^2)$$

$$T = 4\theta \frac{I}{t^2} = \frac{4 \times 10\text{deg} \times (\pi/180\text{deg}) \times (151.1458 \text{ kg} \cdot \text{m}^2)}{(20\text{sec})^2}$$

$$T = 0.2638 \text{ N} \cdot \text{m}$$

Sizing Reaction Wheel

- **Momentum Storage in Reaction Wheel**

- Worst-case Disturbance, 700km, $\frac{1}{4}$ 주기

$$h = (T_{\text{disturbance}}) \times \frac{\text{Orbital Period}}{4} \times (0.707)$$

Orbital Period = 98.8min (700km에서)

$$T_{\text{disturbance}} = 9.3498 \times 10^{-4} \cong 1.0000 \times 10^{-3} \text{N} \cdot \text{m}$$

$$h = (1.0000 \times 10^{-3} \text{N} \cdot \text{m}) \times \left(\frac{98.8 \text{ min}}{4} \right) \times \left(\frac{60 \text{ sec}}{1 \text{ min}} \right) \times (0.707)$$

$$\mathbf{h = 4.1910N \cdot m \cdot s}$$

Sizing Reaction Wheel

- Reaction Wheel의 필요조건

$$T = 0.2638N \cdot m$$

$$h = 4.1910N \cdot m \cdot s$$

Sizing Thruster

- i. Worst-case $T_{\text{disturbance}}$ 를 고려한 Thruster의 최소 추력

$$T_{\text{disturbance}} = 9.3498 \times 10^{-4} \cong 1.0000 \times 10^{-3} \text{N} \cdot \text{m}$$

$$F = \frac{1.0000 \times 10^{-3} \text{N} \cdot \text{m}}{0.5 \text{m}} = 2.0000 \times 10^{-3} \text{N}$$

- ii. Slew Rate(0.05deg/1s)을 고려한 Thruster의 최소 추력

가속을 위해 3초가 걸린다고 가정하고 0.05deg/1s 에 필요한 추력을 구하면

$$\ddot{\theta} = \frac{\dot{\theta}}{t} = \frac{0.05 \text{ deg/sec}}{3} = 0.0167 \text{ deg/sec}^2 = 2.91 \times 10^{-4} \text{ rad/sec}^2$$

$$F = \frac{I \times \ddot{\theta}}{L} = \frac{(151.1458 \text{ kg} \cdot \text{m}^2) \times (2.91 \times 10^{-4} \text{ rad/sec}^2)}{0.5 \text{m}} = 0.0880 \text{N}$$

Sizing Thruster

iii. Momentum Dumping을 위해 필요한 최소 추력

5초의 시간동안 앞서 구한 h값을 Dumping하기위한 추력을 구하면

$$F = \frac{h}{L_t}$$

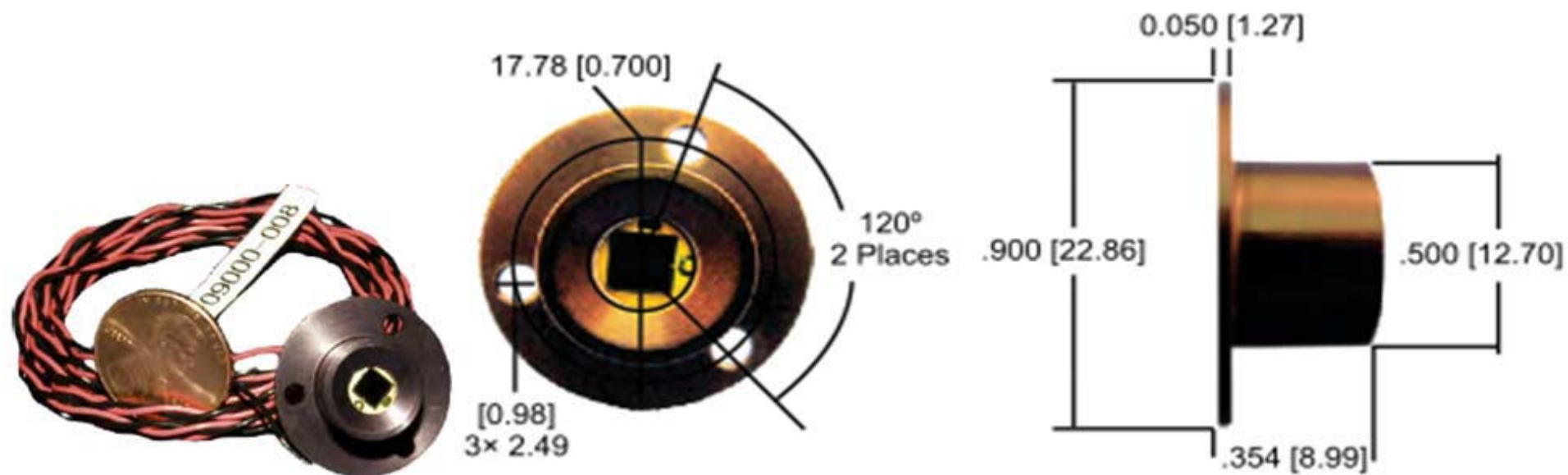
$$F = \frac{4.1910\text{N} \cdot \text{m} \cdot \text{s}}{(0.5\text{m}) \times (5\text{sec})} = 1.6764\text{N}$$

최종값을 구하면 최소 1.6764N 이 된다.

ADCS Over view

	제조사	개수	무게	전력	가용온도
Coarse Sun Sensor	Space Micro	8	10g(80g)	0W	-40 ~ 93°C
Gyro	Northrop Grumman	1	7.1kg	43W	-55 ~ +85°C
Magnetometer	Watson Industries, Inc.	1	90g	0.4W	-50 ~ +80°C
Star Tracker	Jena Optronik	1	1.98kg	10W	-30 ~ +60°C
Reaction Wheel	Honeywell	4	6kg(24kg)	22W	-30 ~ +70°C
Thruster	Aerojet	12	0.33kg (3.96kg)	8.25W	-30 ~ +60°C

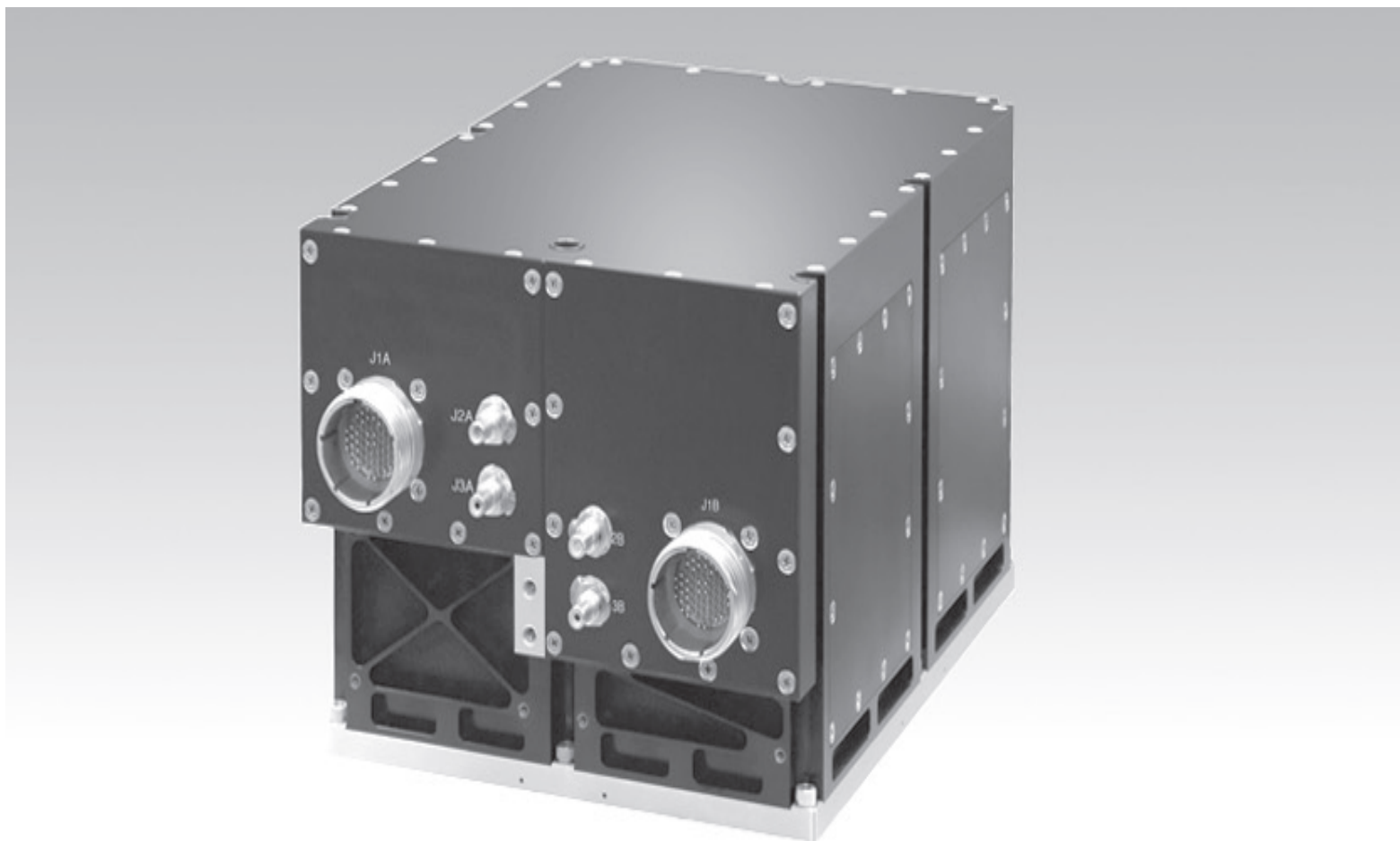
Coarse Sun Sensor



Coarse Sun Sensor

Data & Specifications of Coarse Sun Sensor	
모델	CSS-01
제조사	Space Micro
무게	10g
요구전력	0W
Field of View	120° full-angle circular field of view
Accuracy	5° of 1-axis knowledge
Vibration Test	14.1 grms
Shock Test	60g protoqual
Housing Diameter	1.27cm
Flange Diameter	2.286cm
Sensor Height	0.899cm

Gyro

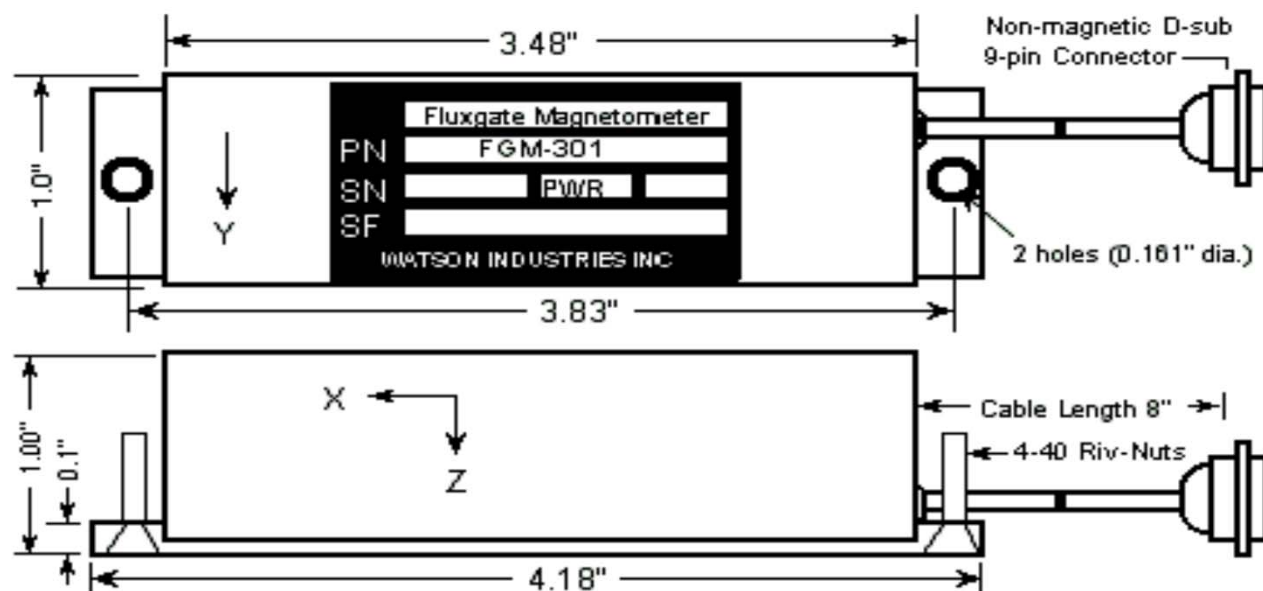


COSMETIC

Gyro

Data & Specifications of Gyro	
제조사	Northrop Grumman
모델	Scalable SIRU
무게	7.1kg
요구전력	43W
가용온도	-55°C ~ +85°C
Ps(15 years continuous operation)	>0.997
Gyro Bias Stability	<0.0015°/hr
Noise Equivalent	<3 arc sec pk-pk
Angle Random Walk	<0.0015°/hr
Scale Factor Short Term Stability	<5 ppm
Random Vibration	18.56g rms
Shock	3000g peak

Magnetometer

**COSMETIC**

Magnetometer

Data & Specifications of Magnetometer	
제조사	Watson Industries, Inc.
모델	FGM-301
무게	90g
요구전력	0.4W
가용온도	-50°C ~ +80°C
Magnetic Range	X, Y, Z $\pm 70,000$ nTesla
Scale Factor	± 200 mGauss/V ± 3.5 V Output
Scale Factor Accuracy	2%
Bias	X, Y, Z < 5 mGauss
Bandwidth	30 Hz
Vibration Survival	4g rms 100 Hz to 1 kHz
Size Including Mounting Flanges	2.5 x 10.6 x 2.5 (cm)

Startracker

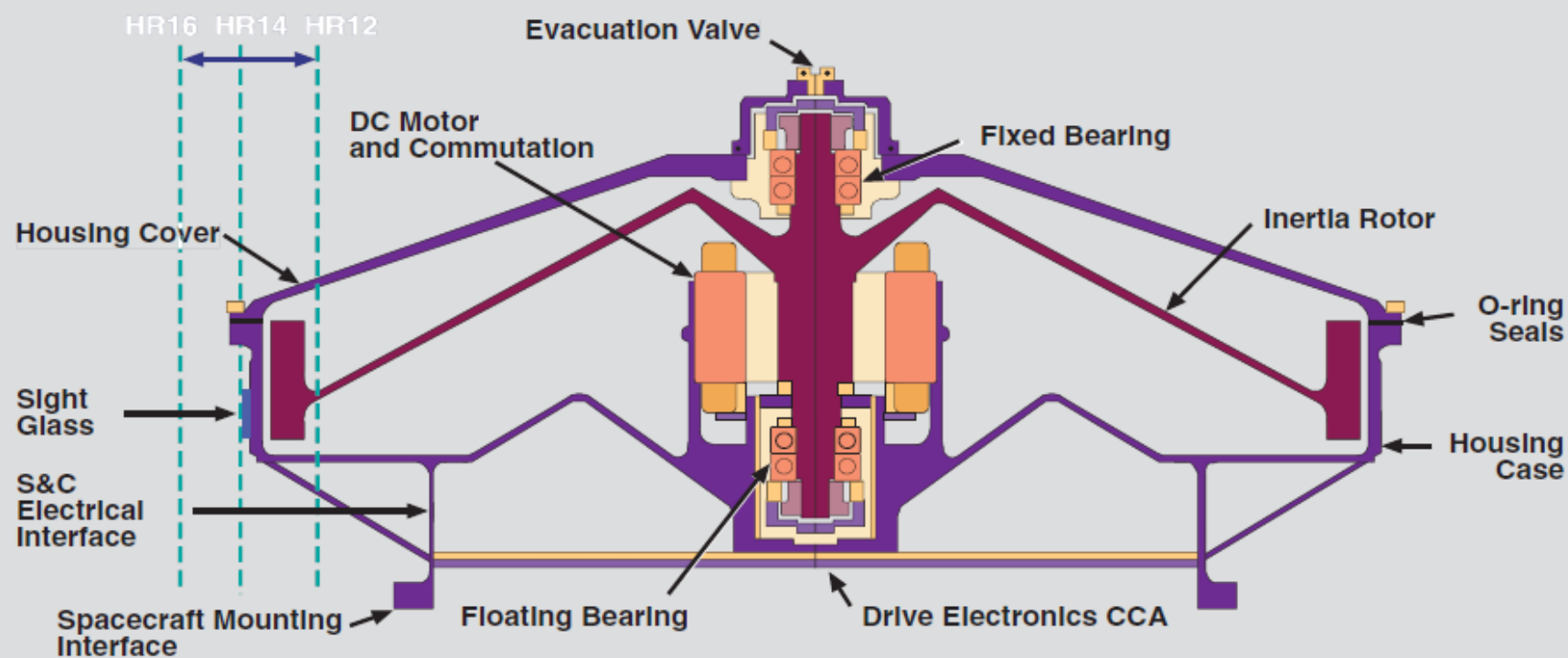


Startracker

Data & Specifications of Startracker	
제조사	Jena Optronik
모델	ASTRO APS for GEO mission
무게	1.98kg
요구전력	<10W
가용온도	-30°C ~ +60°C
Field of View	20° 0 [circular]
Attitude accuracy	< 1 arcsec [1 σ] xy-axes
Attitude accuracy	< 5 arcsec xyz-axes
Slew rate Sampling time	0.3° deg/s [full performance]
Sensitivity Straight light	5.8 mi G0-ref. star, EOL

Reaction Wheel

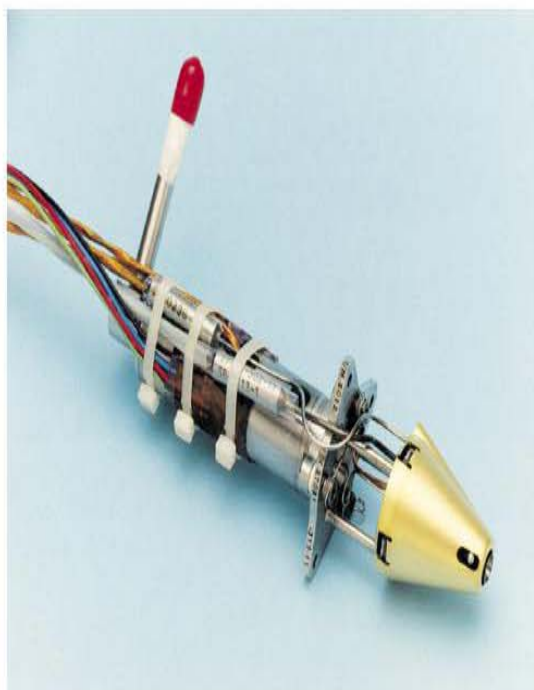
Cross Sectional View



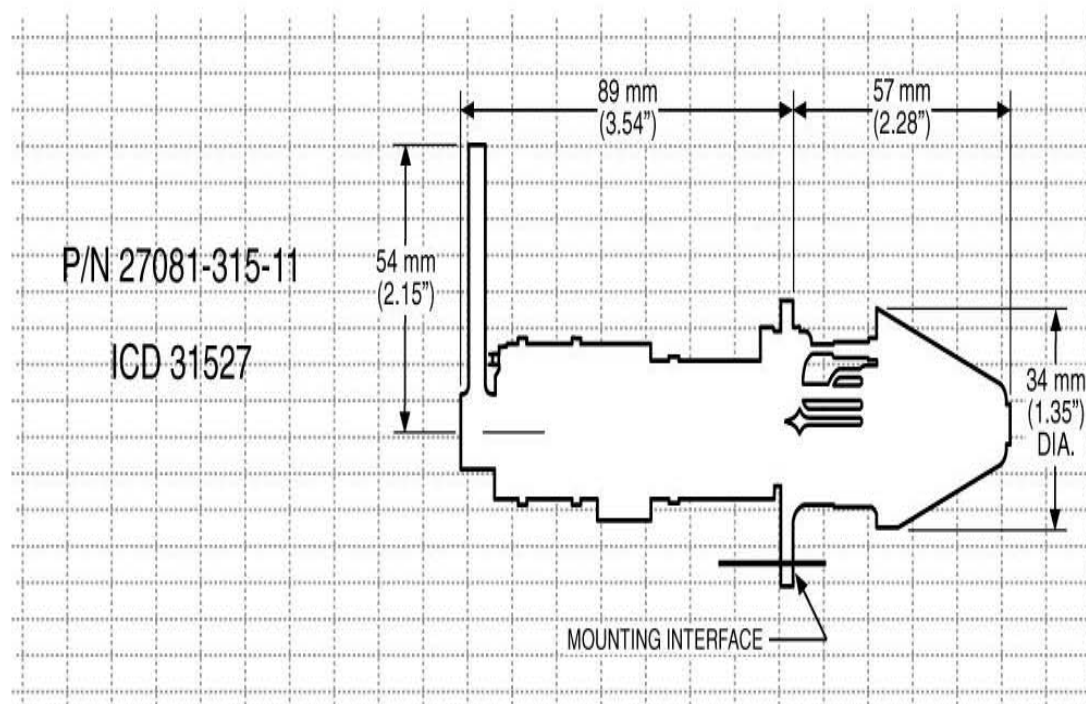
Reaction Wheel

Data & Specifications of Reaction Wheel	
제조사	Honeywell
모델	HR12
무게	6kg
요구전력	<22W
가용온도	-30°C ~ +70°C
Momentum	12Nms
- Reactio torque(nominal)	0.2 Nm
- Reactio torque(Extended)	up to 0.4 Nm (at 3000rpm)
Peak Power	< 105W
Steady State Power(at 6000rpm)	< 22W (typical)
Wheel Outline (Height X Width)	159mm X 316mm

Thruster



4109-1



Thruster

Data & Specifications of Thruster	
제조사	Aerojet
모델	MR-111C 4N
무게	0.33kg
요구전력	8.25W
가용온도	-30°C ~ +60°C
Propellant	Hydrozine
Thrust (Steady State)	5.3N ~ 1.3N
Flow Rate	2.4 ~ 0.6g/sec
Valve	Dual Seat
Total Impulse	260,000Nsec
Total Pulses	420,000

GUIDANCE, NAVIGATION CONTROL & PROPULSION SYSTEM

2014년 12월 11일

Yonsei Asteroid Hit Observation (YAHO) team

이정아



연세대학교 천문우주학과
DEPARTMENT OF ASTRONOMY

Contents

1. 위성의 궤도

2. 위성의 궤도 결정

3. 위성의 궤도 제어

4. 추력계



1. 위성의 궤도

(1) 위성의 초기 궤도



지구 중심 궤도

고도: 700km

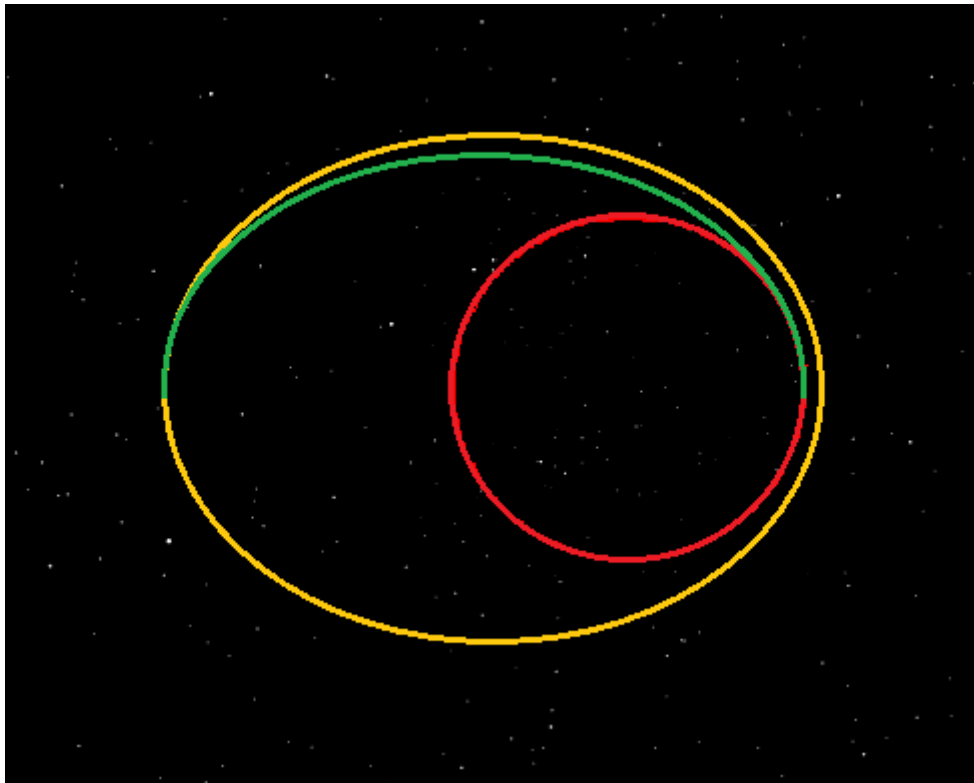
이심률: 0 (원궤도)

궤도 기울기: 0 deg (적도면 공전)

위성의 공전 주기: 1.6459시간

1. 위성의 궤도

(2) 위성의 천이궤도



태양 중심 궤도

장반경: 1.635AU

이심률: 0.388379 (원궤도)

궤도 기울기: 0 deg (황도면)

궤도 천이 시간 = 공전 궤도/2
= 381.8일

1. 위성의 궤도

(3) 위성의 운용 궤도



소행성 중심 궤도

고도: (소행성의 무게중심으로부터)

15km - 100km

이심률: 0 (원궤도)

궤도 기울기: 0 deg

(소행성 계의 공전궤도면과 동일)

위성의 공전 주기: 22.5일(15km)

388일 (100km)



2. 위성의 궤도 결정

(1) Pointing, Mapping 요구 조건

In-Track, Cross-Track: 해당 고도에서 Field Of View 0.1deg안에 secondary asteroid들어오도록 유지

Radial: secondary asteroid 의 1.5m 크기의 크레이터를 분해할 수 있도록 유지

	15km	100km
In-Track	0.0072km	0.002km
Cross-Track	0.0072km	0.002km
Radial	2km	1km



2. 위성의 궤도 결정

(2) 위성의 궤도 결정 방법

방법: Definitive Orbit Determination

궤도 관측 자료	관측 간격	궤도 결정 방법	제어 방법
Deep Space Networks	3일마다 교신	후처리 OD	지상국 제어
Laser Altimeter	10초/10분	후처리 OD	자동화 제어

3. 위성의 궤도 제어

KICK MOTOR

<Orbit Control>

천이 궤도 진입

$$\Delta V = 4.3578 \text{ km/s}$$

최종 궤도 진입

$$\Delta V = 0.0535 \text{ km/s}$$

고도 변화 (15km → 100 km)

$$\Delta V = 0.02461 \text{ m/s}$$

<Orbit Maintenance>

East/West Station Keeping

$$\Delta V = 3 \sim 6 \text{ m/s per year}$$

North/South Station Keeping

$$\Delta V = 45 \sim 55 \text{ m/s per year}$$

MONOPELLANT THRUSTERS



4. 추력계

- 연료

Kick Motor에서 쓰일 연료

	ΔV	연료량
천이 궤도 진입	4.3578 km/s	531.9818 kg
Margin(15%)		79.7973 kg
Residual(2%)		12.2355 kg
Total Propellant		624.0146 kg

4. 추력계

- 연료

Monopropellant Thrusters에서 쓰일 연료

	ΔV	연료량
최종 궤도 진입	0.0535 km/s	7.7248 kg
고도 변화	0.02461 m/s	0.0033kg
E/W 궤도 유지(3년)	0.018 km/s	2.4159kg
S/N 궤도 유지(3년)	0.135 km/s	18.6004kg
자세 제어		30kg
Nominal Propellant		58.7444kg
Margin(15%)		8.8117kg
Residual(2%)		1.3511kg
Total Propellant		68.9072kg

4. 추력계 - 탱크

EPDM (ASTRIUM)

Tank Net Volume (L)	104
Max. Propellant Volume (L)	78
Propellant	Hydrazine
Pressurant Gas	Helium/Nitrogene
Tank Mass (kg)	6.4
Diameter (mm)	586



4. 추력계

- 추력기

RL-10B-2 (Pratt & Whitney Rocketdyne)

Solid Rocket	
Thrust(N)	110,000
specific impulse(sec)	462
Dry mass(kg)	277
Propellant mass (kg)	16823.74
Diameter (m)	2.13
Length (m)	4.14



4. 추력계

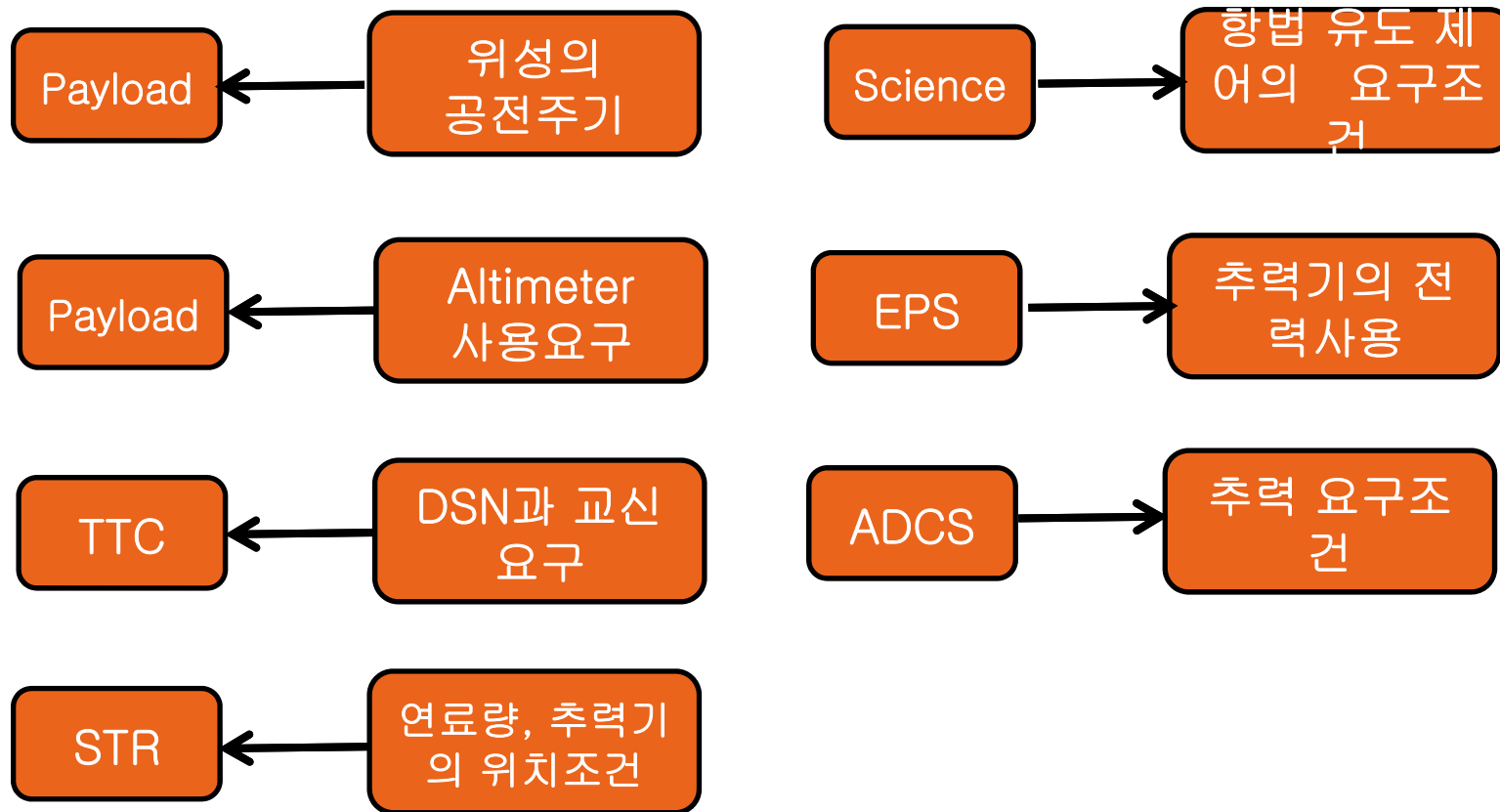
- 추력기

MR-111C (Aerojet) - 12개

Monopropellant Thruster	
Thrust(N)	1.3-5.3N
Specific impulse (sec)	229
Total impulse (N*sec)	260,000N-sec
Propellants	Hydrazine
Exit diameter (cm)	0.19
Length (cm)	16.9
Mass (kg)	0.33



서브시스템간의 상호작용



CNDH & TTC

2014.12.11

YAHN

최 인 영

목차

1. CNDH

2. System and Subsystem Modes

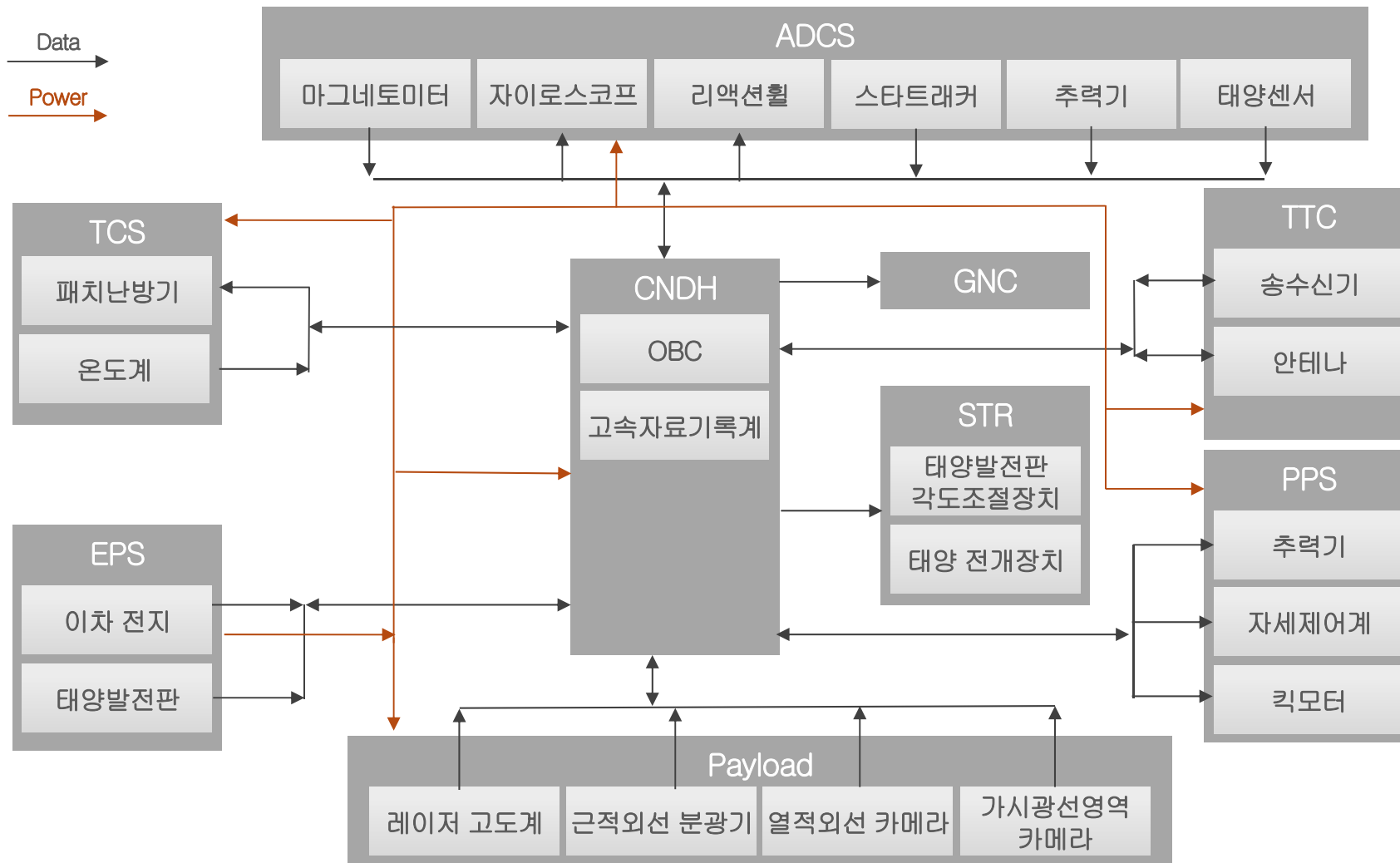
3. TTC

CNDH: Requirements and Constraints

Onboard Computers

- Processing systems
 - Processor performance
 - Failure-safe: Fully redundant
 - Processing resources for subsystems
 - Surveillance and reconfiguration functions
 - Standard interfaces
- Data storage systems
 - Capacity is determined by Payload and Housekeeping data

CNDH: Block Diagram



OBC: Saab Ericsson Space Spacecraft Management Unit

- Highly Reliable using internal redundancy and many heritages
- Support standard interfaces
- Advanced Hardware



General Interfaces	
Digital inputs	0/5V, Relay contact sense, 16-bit serial
Analog inputs	0–5V, –5V to +5V, thermistor
Pulse commands	28V with programmable pulse width, 5V CMOS, 16-bit serial

ADCS Interfaces
Magnetometer interfaces
Gyro interfaces
Reaction wheel interfaces
Star tracker interfaces
Thruster control interfaces
Sun sensor interfaces

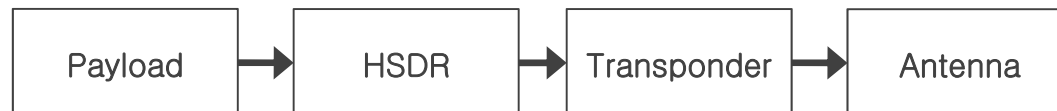
Typical Properties	
Power consumption	Avg. < 40W, Peak < 60W
Mass	16 kg
Dimensions	420 x 270 x 276 mm
Reliability	15-year

Payload Interfaces
1553 Data buses
40 Mbps SpaceWire links
20 Mbps RS-422 Synchronous serial links
1.5 Mbaud UART links RS-422 or RS-485
RS-422 Synch pulses fixed/programmable
Heritages
Herschel, Planck, Pleiades, Aeolus, Galileosat, GAIA, Sentinel-2 and 3, Small GEO, ExoMars Rover

HSDR: SURREY Satellite technology

HSDR (High Speed Data Recorder)

- On-board storage of large volumes of data
- Handle high data acquisition rates from payload side for science data storage
- Fast data output channels to transponders for downlink to ground via X-band or Ka-band

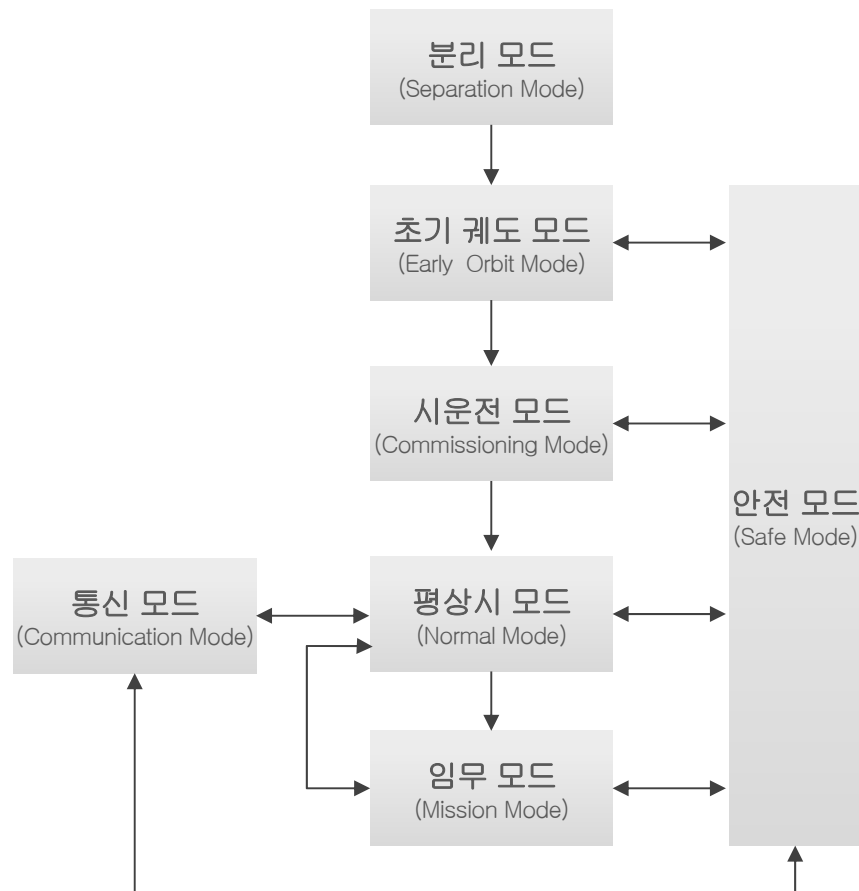


Performance Properties	
FPGA	Xilinx Virtex-4
Storage capacity	16 Gbytes DDR2
Data interfaces	20LVDS 150Mbps Inputs 5 serDes 1Gbps Outputs 16 LVDS 150 Mbps
Other inputs	Pulse-per-second synchronisation
TTC interface	LVDS, CAN, RS-422

Physical Properties	
Dimensions	320x170x55 mm
Power	3.5W standby, 15W peak
Mass	1 kg
Radiation Tolerance	5k Rad(Si)
Lifetime	7 years

Heritage	
Delimos-1, UK-DMC-2, NigeriaSat-2, ExactView-1, TechDemoSat-1, KazEOSat-2, DMC-3 constellation, NovaSAR-S	

System and Subsystem Modes



분리 모드 (Separation Mode)

- 안테나의 전개 수행위해 전개 장치에 명령 전송

초기 궤도 모드 (Early Orbit Mode)

- 전개가 모두 정상적으로 이루어지면 위성의 상태 정보 수집과 송신을 시작
- 궤도 조정 및 궤도 접근을 위한 궤도 안정성 확보

시운전 모드 (Commissioning Mode)

- 초기 궤도 모드에서 통신 종료 명령 수신하면 시운전 모드 진입
- 탑재체가 정상적으로 작동하는지 확인
- 전력계 모듈로부터 각 하드웨어에 공급하는 전류와 전압 정상작동판단

평상시 모드 (Normal Mode)

- 상대정보를 수집하며 통신 시작 확인
- 통신 모드로 진입시 전력이 충분한지 확인 후 진입

임무 모드 (Mission Mode)

- 소행성 관측시의 모드
- 탐채체를 이용하여 임무 수행: 소행성에 15km 접근하여 관측 후 100km까지 떨어진 후 충돌 관측
- 전력이 35% 미만이면 평상시 모드로 진입

통신 모드 (Communication Mode)

- 지상국과 통신 수행: 지구 중심 지향을 수행하고, Transponder에 송신 명령 전달 및 지상으로부터 명령 지속 확인
- 관측 결과 전송

안전 모드 (Safe Mode)

- 전력 부족, 통신 실패, 등 비상 상황에 대비한 대기모드
- 다른 하드웨어의 전원 제어 및 정상적으로 운영될 수 있도록 이삭을

TTC: Requirements and Constraints

Long distance

- NASA Deep-Space Network
Compatible
- Communication Performance
- Signal Latency

Frequency

- X-band (8 to 12 GHz)
 - Uplink: Commands
 - Downlink: Real-time telemetry,
Playback telemetry
- Ka-band (27 to 40 GHz)
 - Downlink: Transmission of
payload mission product data,
system complementary data,
Housekeeping data

Why use Ka-band?

- x4 Increase the data return to the Earth
- Reduce the S/C Antenna
- Reduce the plasma effects
- Increase the SNR in the carrier tracking loop

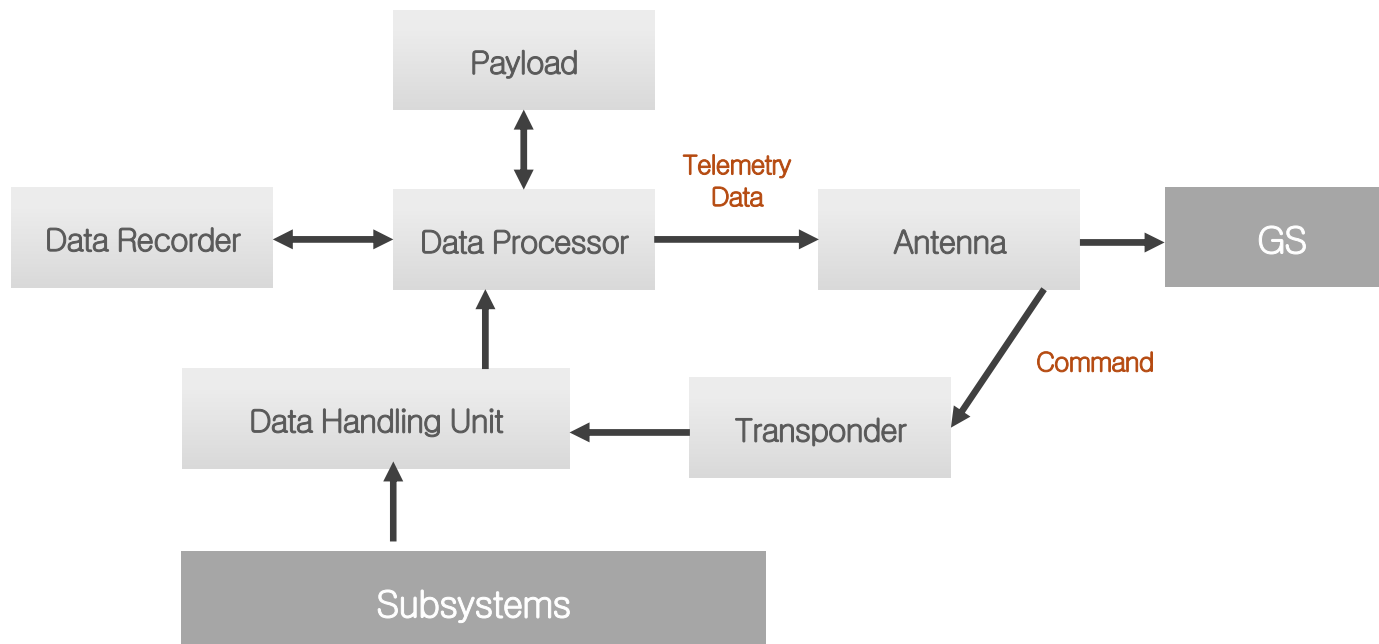
TTC: Subsystems

Transponder

- NASA Deep-Space Network Compatible
- Qty 2 for redundancy

Antenna

- Minimize side lobe
- High Gain



Transponder: GENERAL DYNAMICS

SDST (Small Deep Space Transponder)

- NASA Deep-Space Network Compatible
- X-Band Receiver, X and Ka-Band Exciters
- MIL-STD-1553 Interface: Standard, Low power
- Radiation and SEU Resistant

Heritage

- Mars Exploration Rover



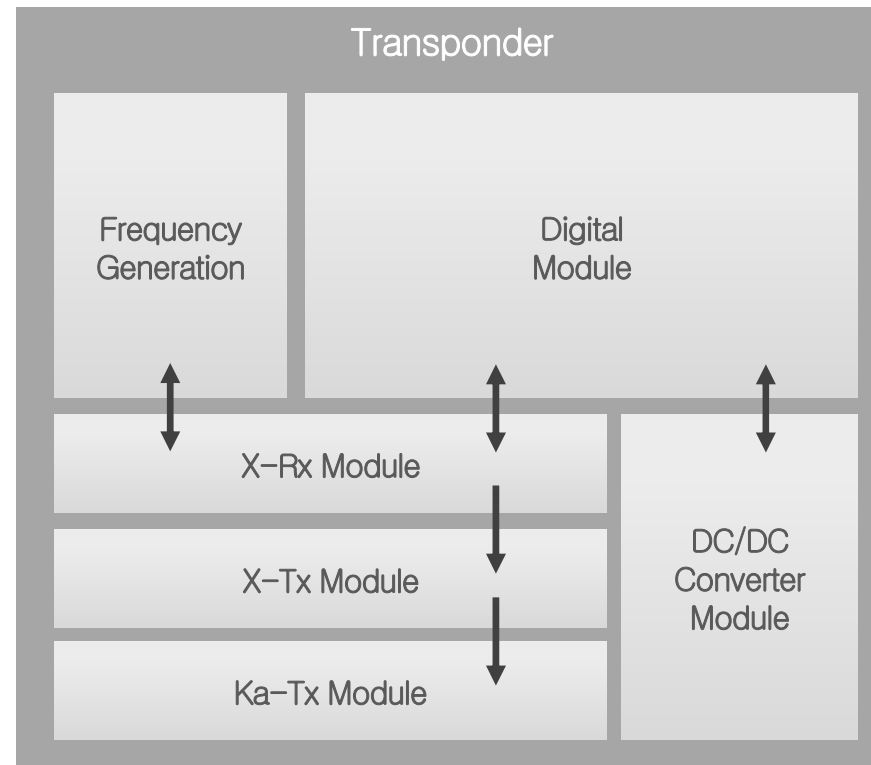
Performance Properties		
X-Band	U/L	7.145 – 7.235 GHz
	D/L	8.400 – 8.500 GHz
	TX/RX Ratio	880/749
Ka-Band	D/L	31.800 – 32.300 GHz
	TX/X-Band RX Ratio	3360/749
Noise Figure		2.1 dB @25°C, 3.2 dB
X-Band Receiver Data Rates		7.81 – 4000 bps (PM/PSK, NRZ)
Carrier Tracking Threshold		–158 dBm

Physical Properties		
Dimensions	181x166x114 mm	
Power	Receiver	12.5 W
	Rx + X-Tx	15.8 W
	Rx + X-Tx + Ka-Tx	19.5 W
Mass	3.2 kg	
Radiation Dose	50 krads (Si)	
Operating Temperature	–40 to +60°C	

Transponder: GENERAL DYNAMICS

Functions

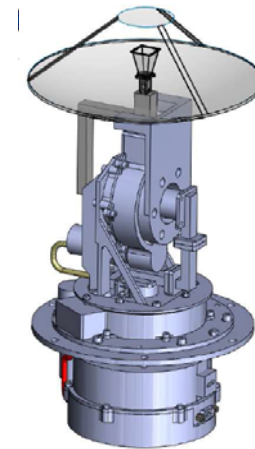
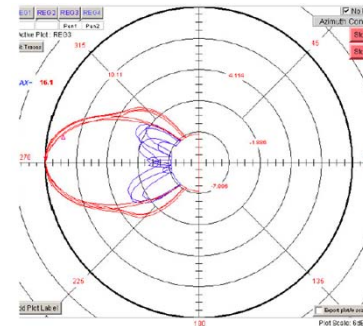
- Command
 - Code Block Processing
 - Data Demodulation
 - H/W Command decoding
- Telemetry
 - Modulation Encoder: Reed–Solomon Encoding
 - Uplink carrier tracking, Downlink carrier generation
 - Ultra Stable Oscillator (USO): Provide a reference for generating downlink carriers



Antenna: SURREY Satellite technology

X-Band/Ka-Band Horn Antenna

- Antenna Pointing Mechanism (APM)
- Two-axis steerable mechanism
- Enable transmitter power consumption low
- Enable GS dish size small
- High agile pointing
- Narrow-beam, High-gain Horn type



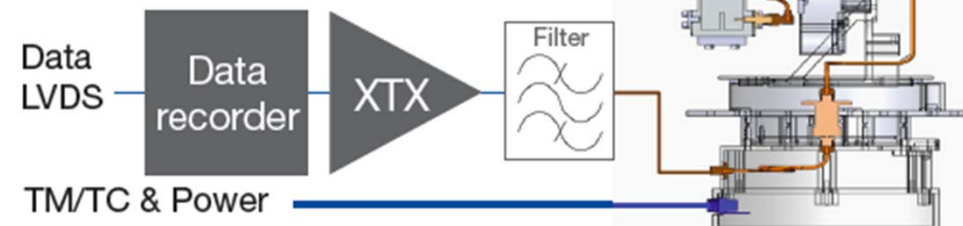
<Ka-band Horn Antenna>

Support High data rate downlinks to Earth and Science missions

- X-APM + XTx: typically deliver 400Mbps
- Ka-APM + KaTx: 500Mbps to 1Gbps

Heritage

- NigeriaSat-2, TechDemoSat-1, Kazakhstan MRES, DMC-3 constellation



<X-band Horn Antenna>

Antenna: SURREY Satellite technology

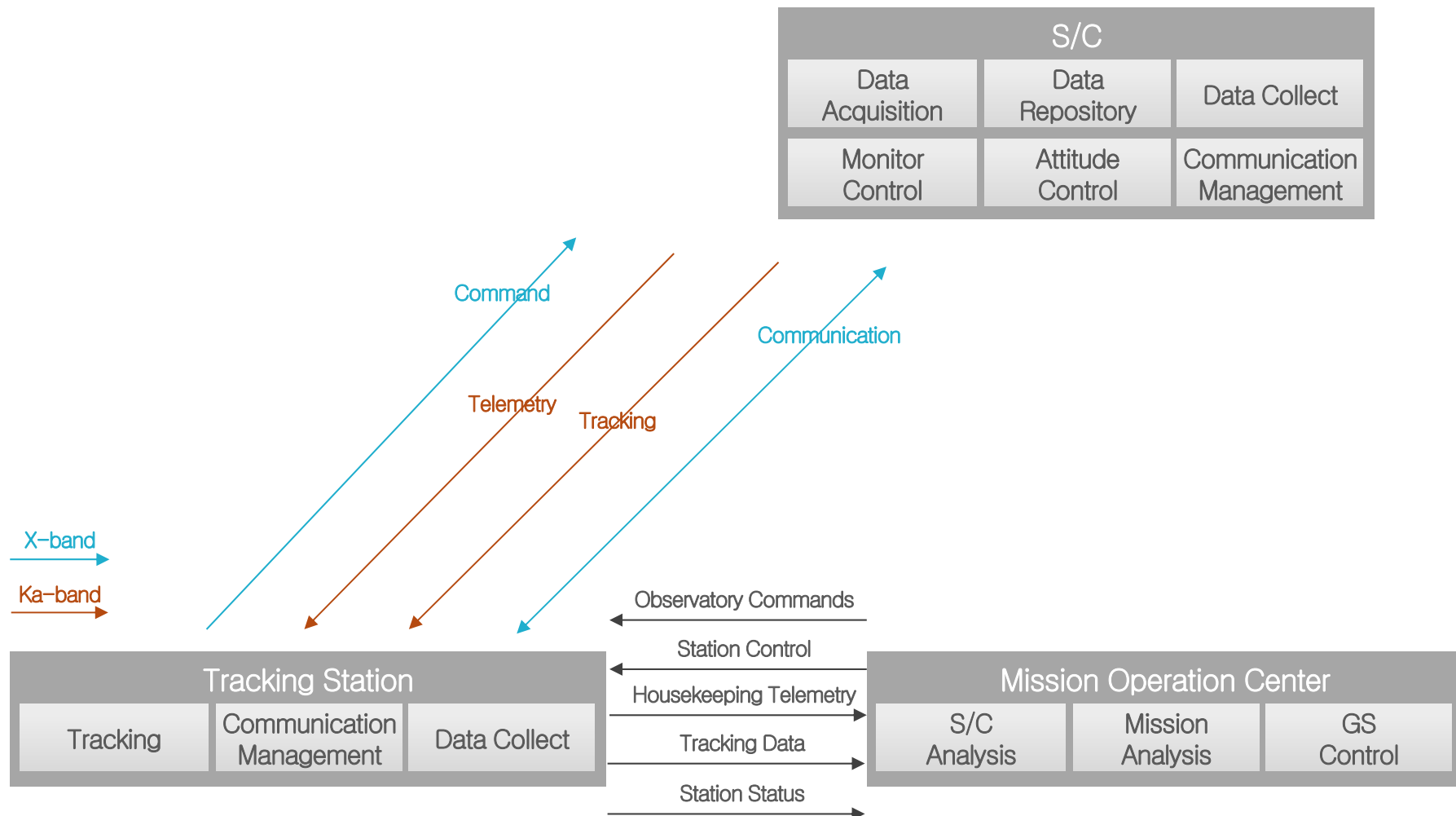
X-Band Horn Antenna

Physical Properties	
Dimensions	Ø274xh300 mm
Power	1.3W standby 3.9W dynamic
Mass	3.3kg
Radiation Tolerance	> 5kRad(Si)
Random vibration	20 G _{rms} in all axis
Lifetime	7 years
Temperature	-40 to +60°C
Performance Properties	
Gain	18 dBiC
Frequency	8.5 GHz
Antenna Axial Ratio	< 3dB
Pointing Accuracy	< 0.25°
Step Size	< 0.024°
TTC Interface	CAN, RS422
Slew Rate	< 20°/s
Slew Range	Azimuth: ±270° Elevation: ±110°

Ka-Band Horn Antenna

Physical Properties	
Dimensions	Ø274xh300 mm
Power	11W
Mass	5 kg
Radiation Tolerance	> 5kRad(Si)
Random vibration	20 G _{rms} in all axis
Lifetime	7 years
Temperature	-40 to +60°C
Performance Properties	
Gain	> 30 dBiC
Frequency	27 GHz
Antenna Axial Ratio	< 3dB
Pointing Accuracy	< 0.25°
Step Size	< 0.024°
TTC Interface	CAN, RS422
Slew Rate	< 20°/s
Slew Range	Azimuth: ±270° Elevation: ±110°

TTC: Block Diagram



Ground System

NASA Deep Space Network

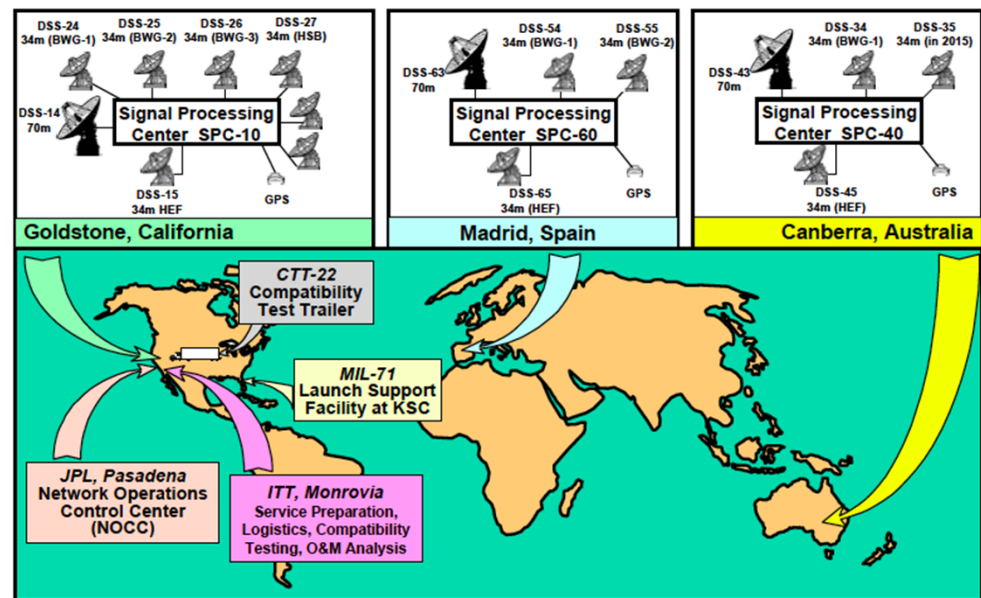
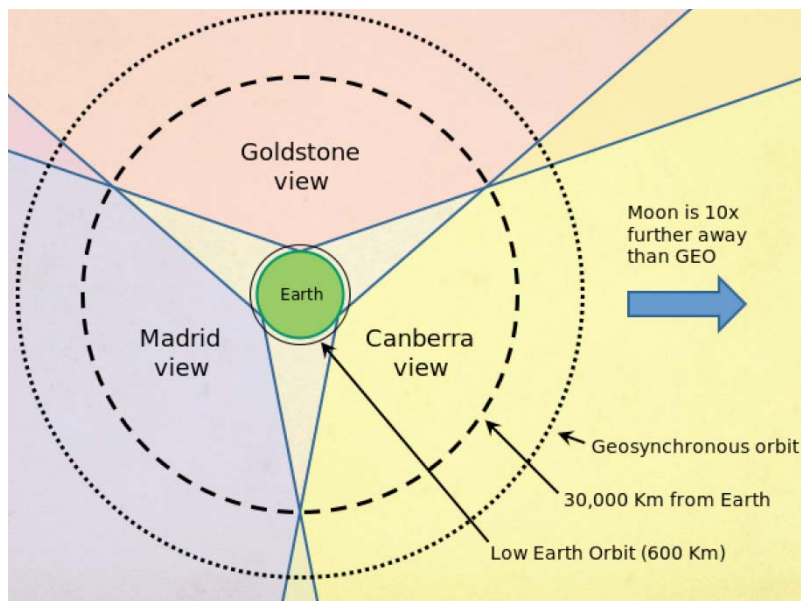
- DSN facilities in California, Madrid, Canberra can cover the whole angles
- No GPS
- Having very large ground antennas

X-band

- 34m antenna
- Data rate (2.6 AU): 1 Mbps

Ka-band

- 34m, 70m antenna
- Data rate (2.6 AU)
 - 34m antenna: 1.5 to 23 Mbps
 - 70m antenna: 54Mbps



감사합니다

전력 부시스템

2014.12.11

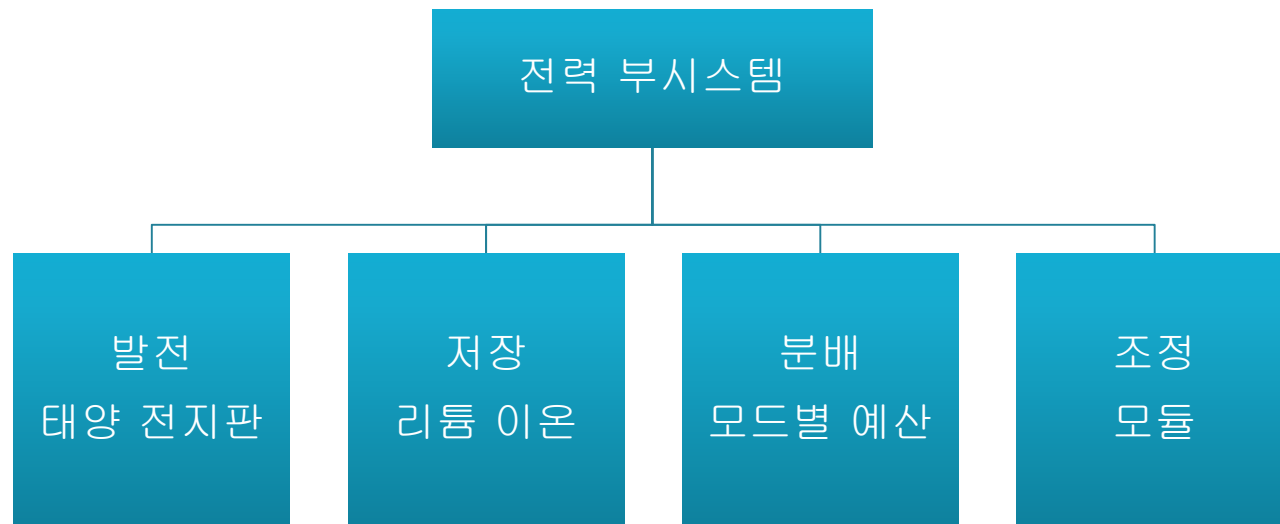
YAHN

방정훈

목차

-
1. 전력 요구/구속조건
 2. 전력 발생 장치
 3. 전력 저장 장치
 4. 전력 분배 및 실패 보호
 5. 전력 세부 예산
-

전력 부시스템의 구성



전력 부시스템 요구 및 구속조건

임무기간 **5년** 동안 위성체에 연속적인 전력을 공급

위성체에 공급되는 전력을 조정하고 분배

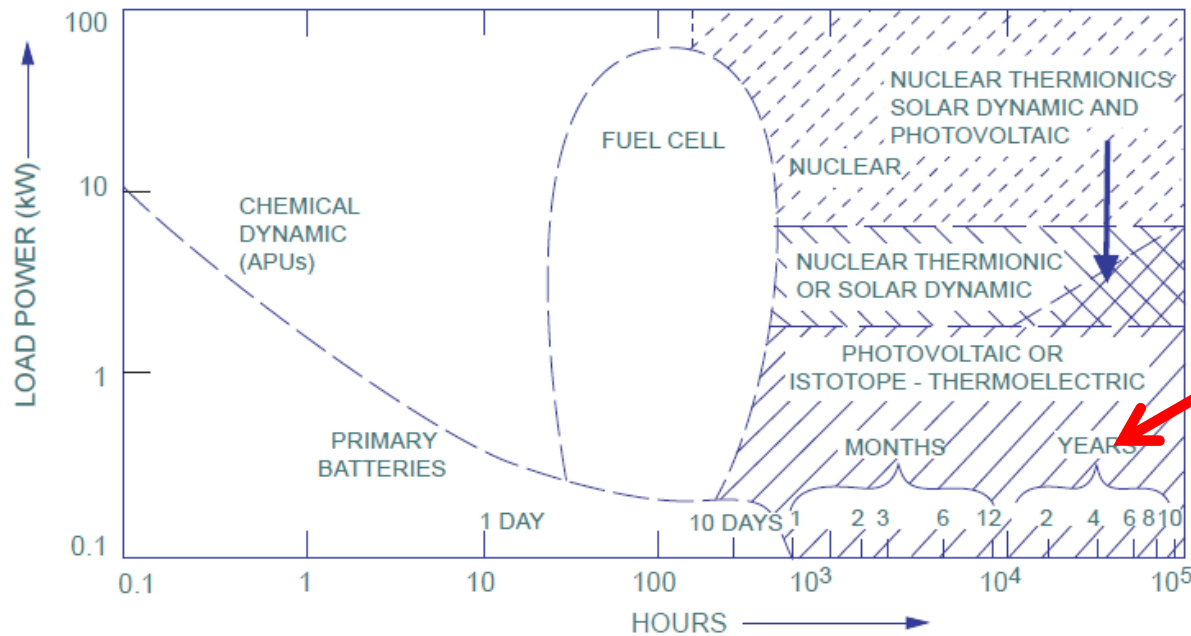
평균 **495W** 및 최대 **732W** 전력 부하가 올 때 지원

필요하다면 전력 공통부분에서 전압 전환 및 안정화 (**DC-DC Conv.**)

명령계와 통신계에 지상국의 조정 / 자가 조정이 가능하도록 전력을 제공 (**항시 40W**)

전력부 고장이 일어날 때 탑재체를 보호 **안전 모드**

전력 부시스템 요구 및 구속조건



Approximate ranges of application of different power sources.

David W. Miller

5년

평균 495W

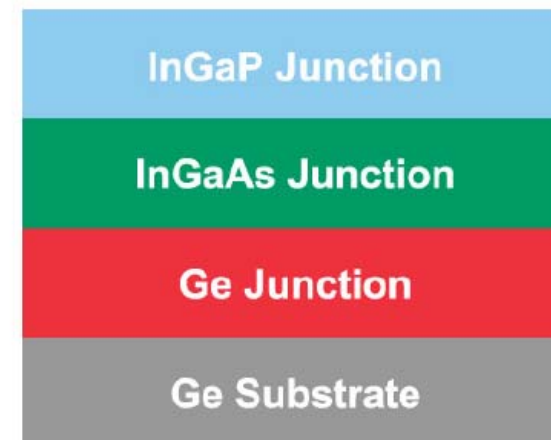
발전 장치

삼중 교차 태양 전지판
(최저 평균 효율 29.5%)

강한 EOL/BOL 유지 비율
(0.97 at $10^{14}\text{e}/\text{cm}^2$)

SQ 결과

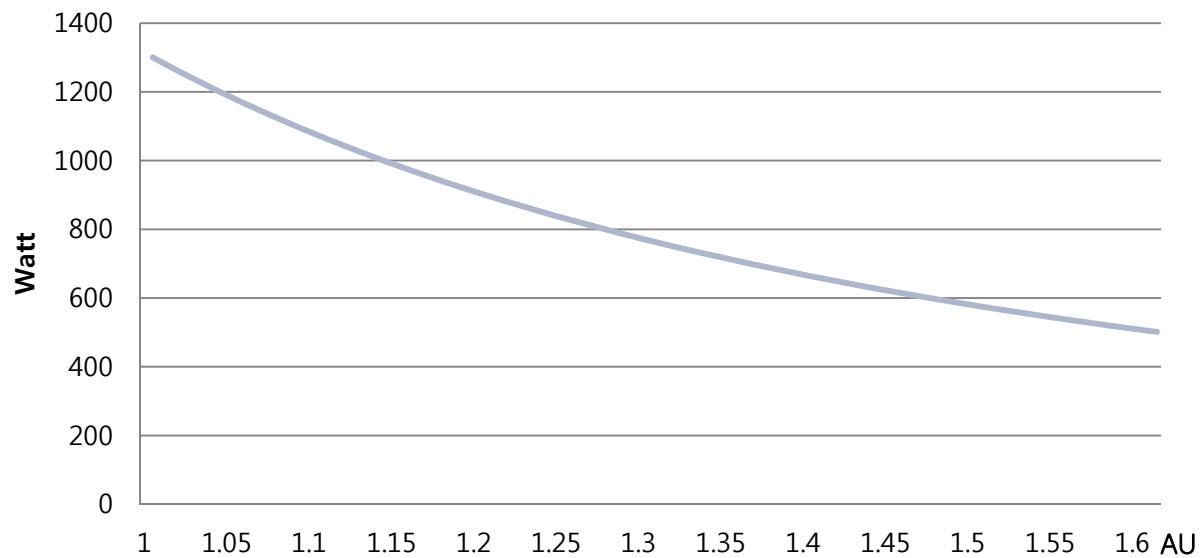
- 열 내구도 실험 2000회; 변화 없음
- 40일 습도 변화 실험 ;변화 없음



©EMCORE Corporation

전력 요구조건

거리 / 발전량



목표

$P_{sa} = 1287 \text{ W}$
(495W의 2.6배)

$P_0 = 301 \text{ W/m}^2$
 $I_d = 0.77$
 $\Theta (\text{SADM 장치}) = 0$

$P_{EOL} = 226.03 \text{ W/m}^2$
(5년 절감 고려)

$A_{sa} = 5.69 \text{ m}^2$
 $M_a = 4.84 \text{ kg}$
(3중 140 μm 두께)



© Surrey Satellite Technology Ltd.

저장 장치

리튬 이온 전지 (Lithium Cobalt Oxide)

<Worst case>

최대 출력 732W (구동시간 2h, DoD80%)

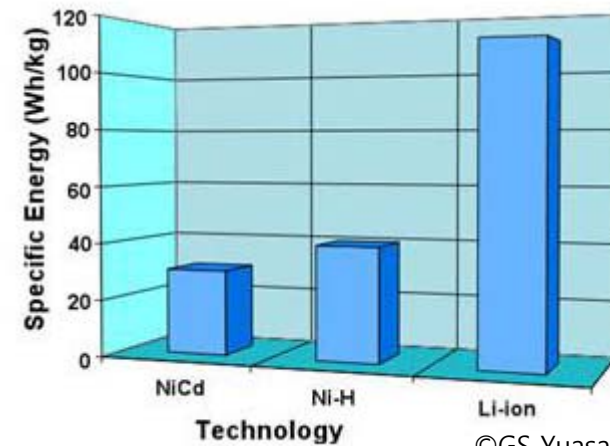
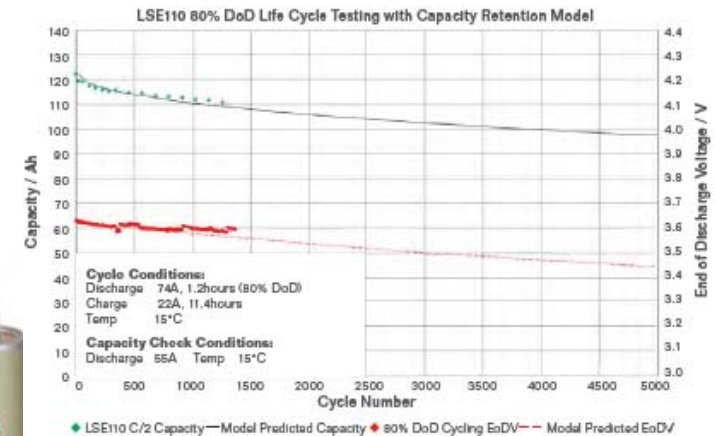
배터리의 개수는 총 8개 (596.4Wh)

<장점>

높은 에너지 밀도 (28.4kg)

변화 없는 EoDV(방전종료전압)

충/방전 횟수가 매우 적음



©GS Yuasa

전력 분배 및 실패 대비

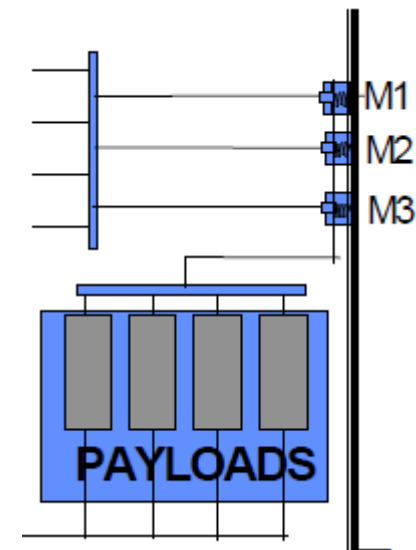
전력 분배 모듈 + DC-DC 변압기

Payload 보호는 마이크로스위치에
On/Off 명령

저장 장치 실패는 여분의 4개 배터리로 대체

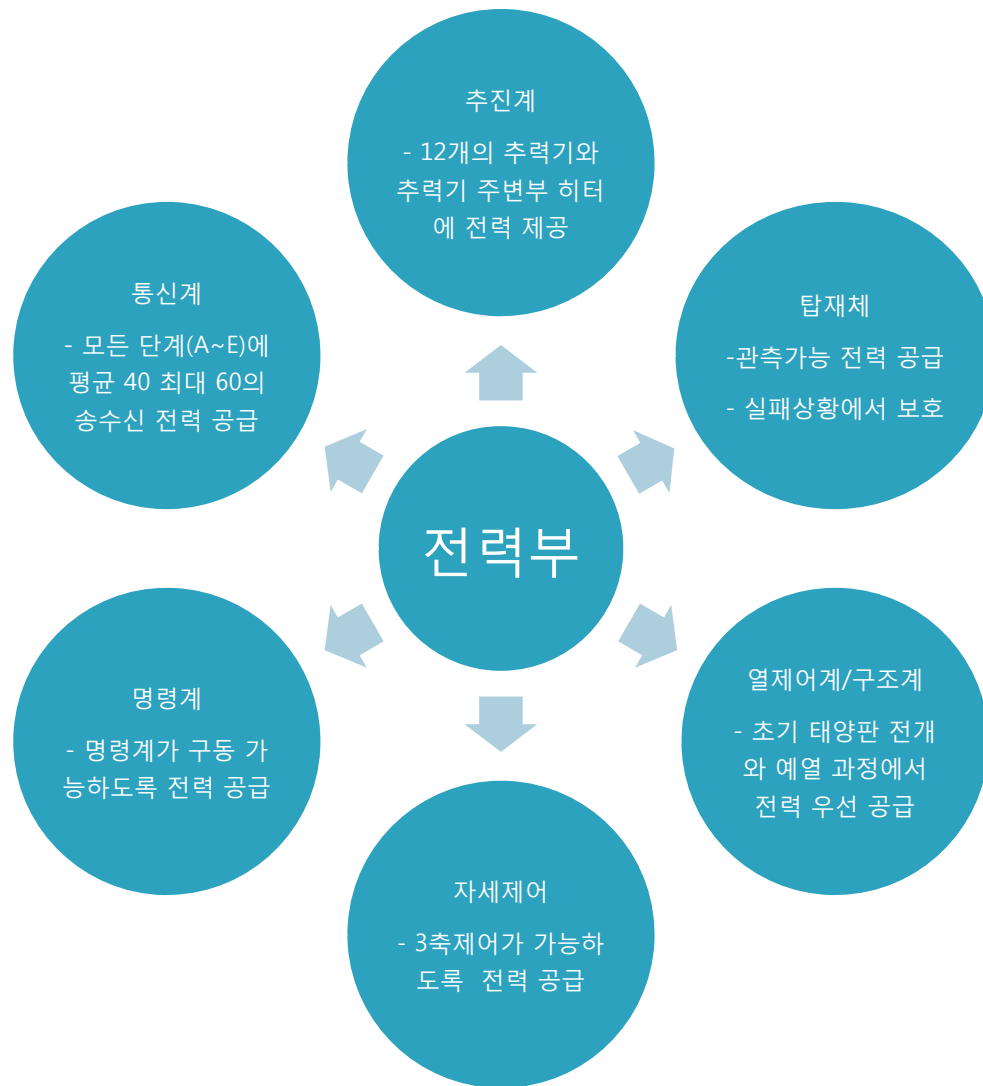


©Clyde Space

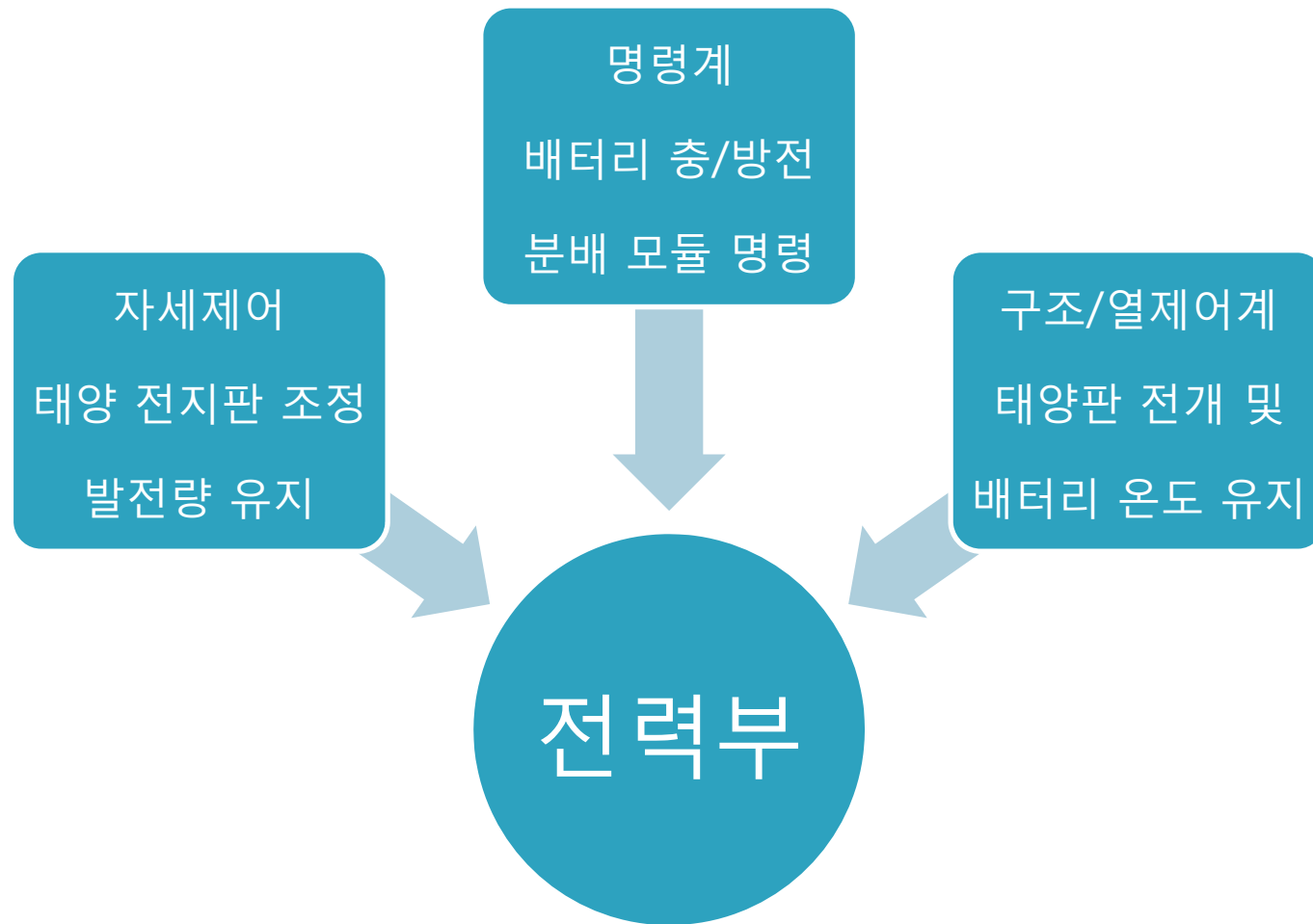


©David Groch

전력 분배 및 상호작용



전력 분배 및 상호작용



전력 단계

A단계 : 시작, 전개

B단계 : 지구 궤도

C단계 : 소행성 천이 궤도

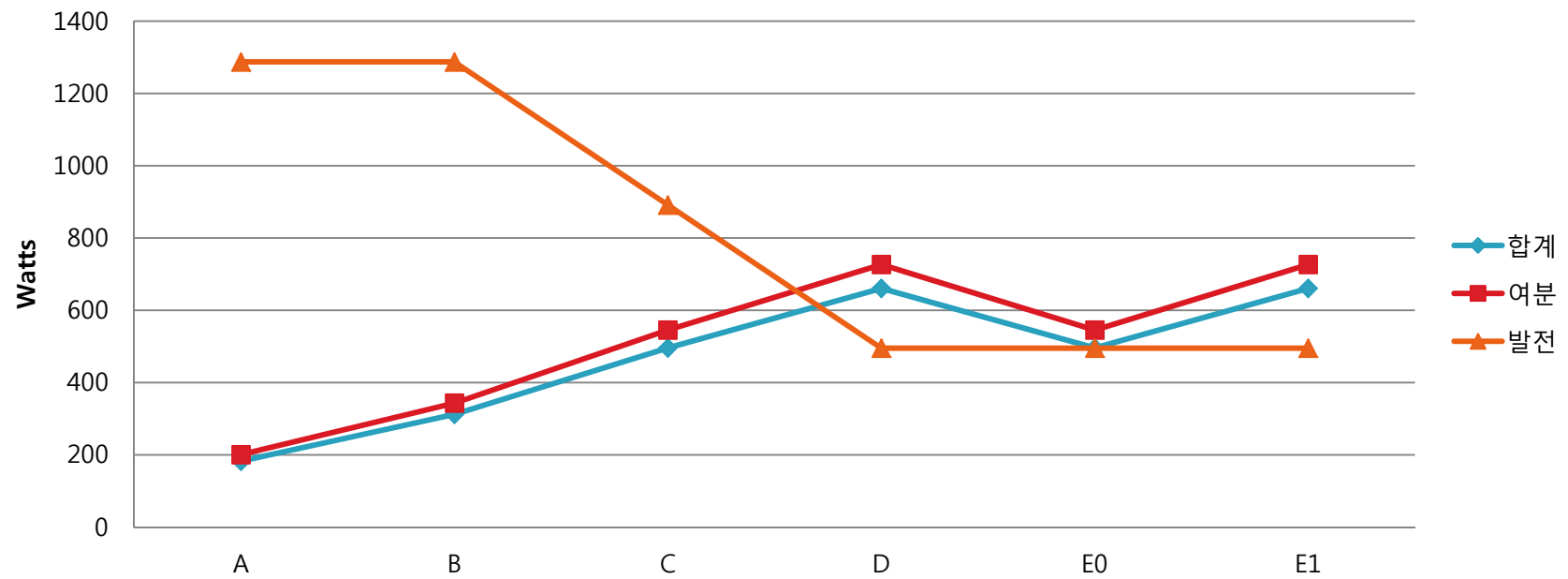
D단계 : 소행성 15km 궤도

E단계 : 충돌 직전 100km 궤도

궤도 진입 상태에서의 YAH0 전력 사용조건							
목차	항목	단계	A	B	C	D	E
1.1	레이저 고도계		X	X	X	O	O
1.2	근적외선 분광기		X	X	X	O	O
1.3	열적외선 카메라		X	X	X	O	O
1.4	가시광선 카메라		X	X	X	O	O
2.1	패치 난방기		O	O	O	O	O
3.1	태양 발전판 각도 조절장치		O	O	O	O	O
3.2	태양 전개장치		O	X	X	X	X
4.1	태양 센서 Space micro社(2)		X	O	O	O	O
4.2	마그네토미터		X	O	O	O	O
4.3	자이로스코프		X	O	O	O	O
4.4	별 센서		X	O	O	O	O
4.5	리액션 휠(3)		X	O	O	O	O
6.1	추력기(12)		X	X	O	O	O
7.1	X-대역, Ka대역 안테나		O	O	O	O	O
7.2	송수신기		O	O	O	O	O
8.1	위성체 관리 모듈		O	O	O	O	O
8.2	고속 자료 기록계		X	O	O	O	O
9.1	태양 발전판		O	O	O	O	O
9.2	이차 전지		O	O	O	O	O
9.3	전선		O	O	O	O	O

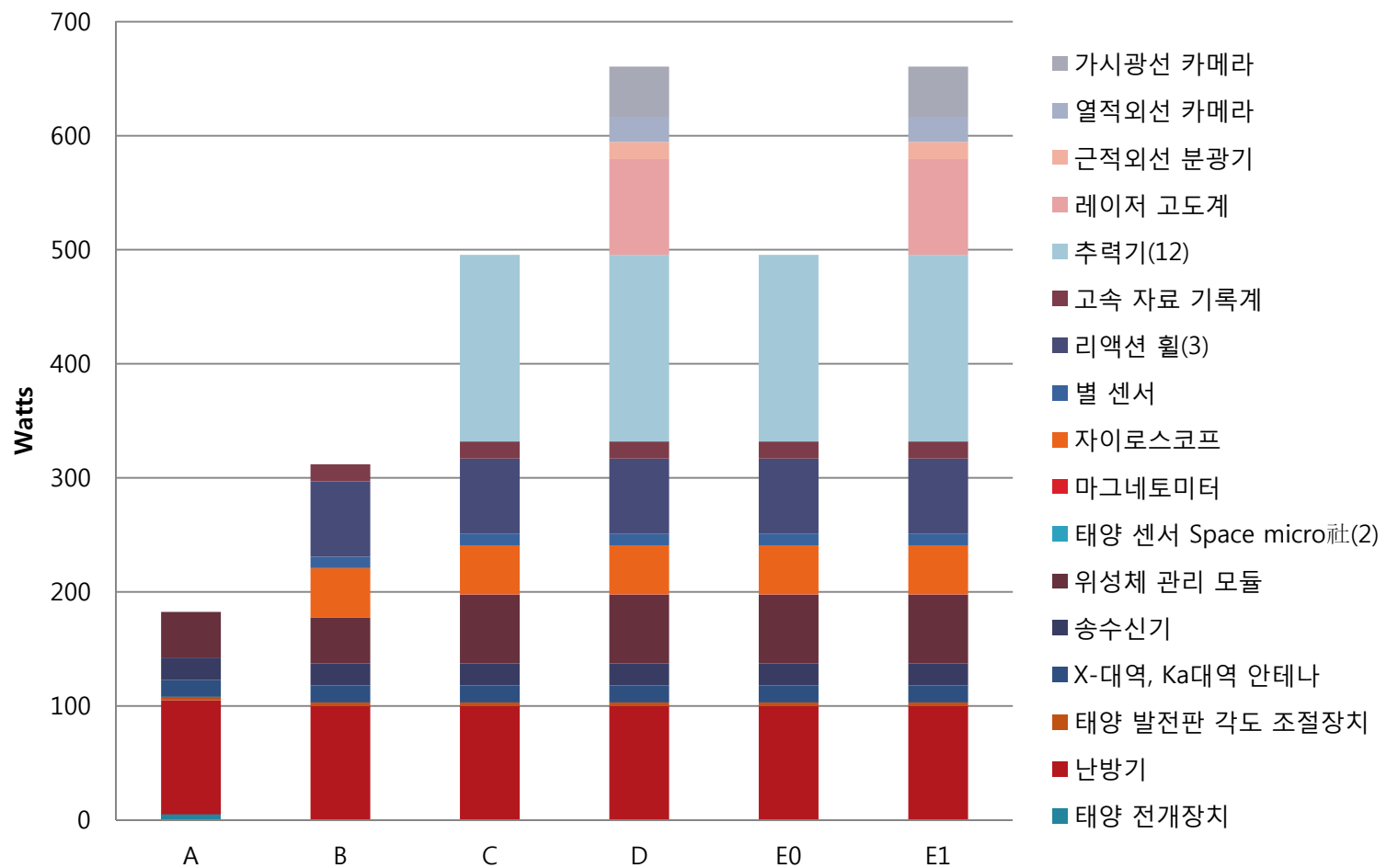
전력 단계

YAHOO의 발전량 / 사용량 변화



전력 단계

각 궤도 진입 상태에서의 YAH0 전력 예산



참고 문헌

1. EPS 기본 및 설계 과정

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Robert Burt, Distributed Electrical Power System in Cubesat Applications, 2011

2. 전력 기기 정보

Solar Array Drive Mechanisms(SADM), Surrey Satellite Technology Ltd. www.sstl.co.uk,

ZTJ Photovoltaic Cell, Emcore corporation, www.emcore.com

Lithium Ion Cells LSE Gen III(145), GS Yuasa Lithium Power, Inc. www.gsyuasa-lp.com

AXALU aluminum wire, Axon cable, www.axon-cable.com

Battery Electronic Unit(BEU), Thales Alenia Space ETCA, www.thalesaleniaspace.com

Power Conditioning and Distribution Unit(PCDU), Surrey Satellite Technology Ltd.

www.sstl.co.uk,

Radhard DC-DC Converters, Clyde Space Ltd, www.clyde-space.com

Thermal Control System Structure & Mechanism System

2014.12.11

YAHN

연세대학교 천문우주학과 이영로

목차

Thermal Control System

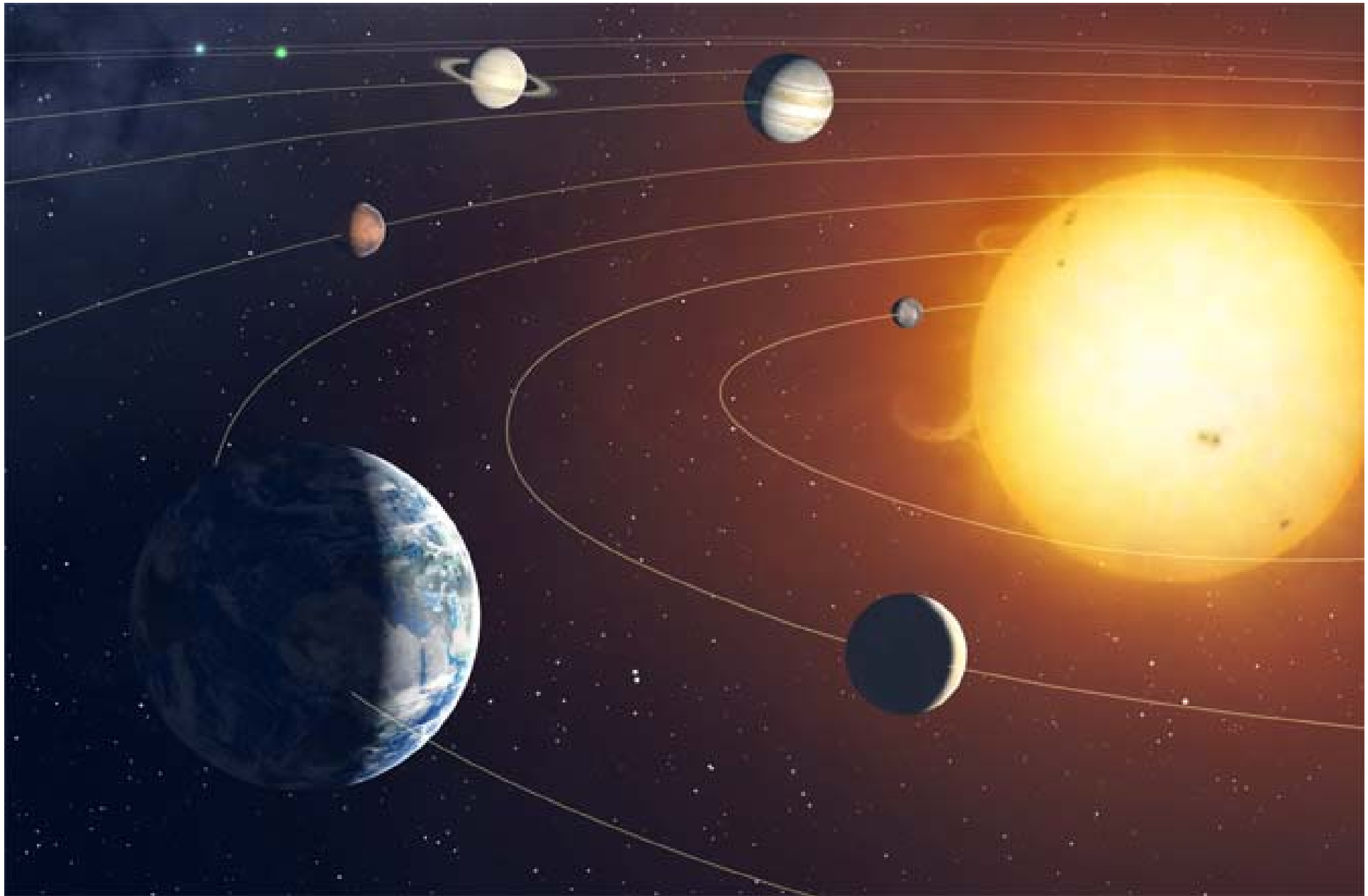
1. Thermal Control System

2. 각 Subsystem의 온도 요구조건

3. 우주 환경에 따른 S/C의 온도

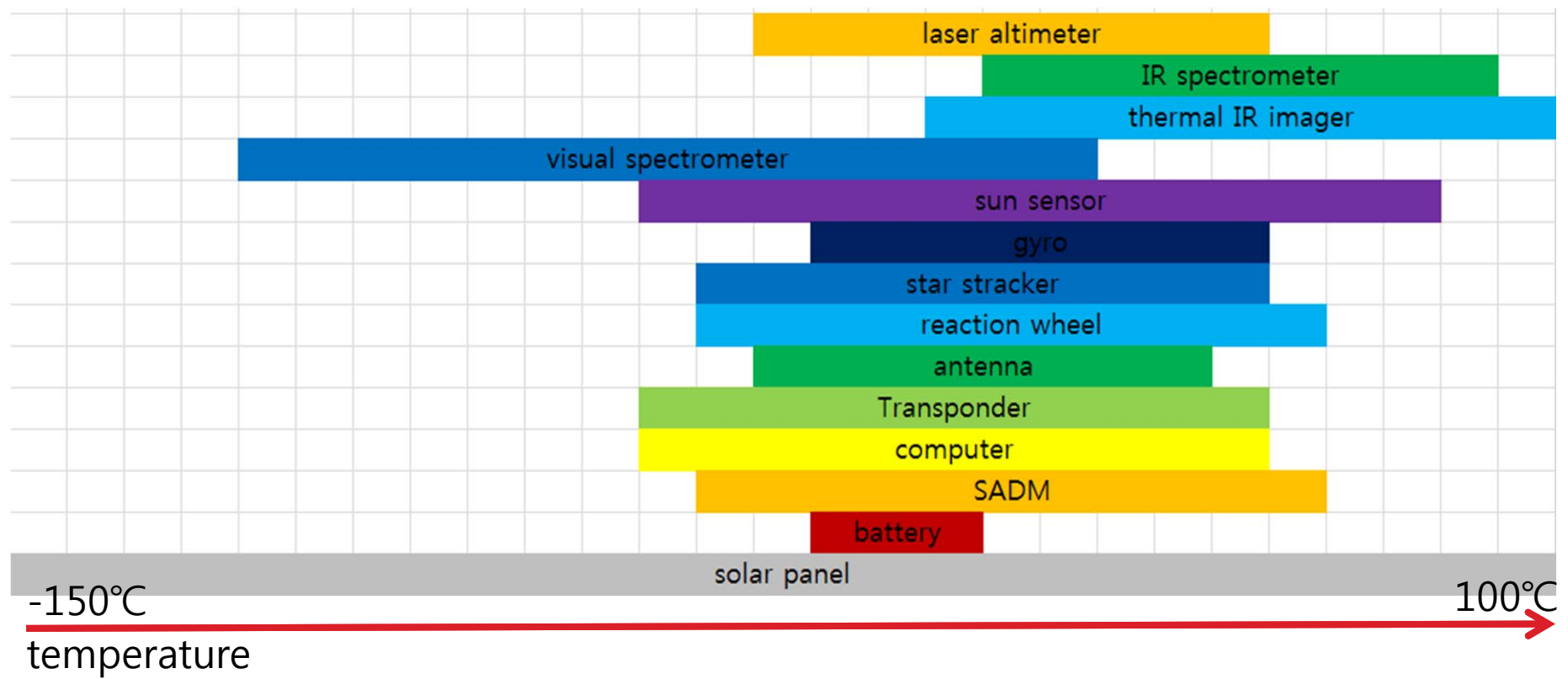
4. YAHOO 의 Thermal Control

1. Thermal Control System



2. 각 Subsystem 의 온도 요구조건

component		temperature range(°C)	
		Operational	Survival
PAYLOAD	laser Altimeter	-20~60	-30~70
	IR spectrometer	20~100	10~110
	Thermal IR imager	12~120	2~130
	Visual spectrometer	-113~27	-123~37
ACDS	Sun sensor	-40~93	-50~103
	Gyro	-10~60	-55~85
	Star tracker	-30~60	-40~70
	Reaction wheel	-30~70	-40~80
TTC	Antenna	-20~50	-30~60
C&DH	Transponder	-40~60	-50~70
	Onboard computer	-40~60	-50~70
EPS	SADM	-30~70	-40~80
	Battery	-10~10	-10~35
	Solar panel	-150~110	-200~130



3. 우주 환경에 따른 S/C의 온도

소행성궤도(AU)		태양flux (W/m ²)
perigee	1.01	1340
apogee	2.28	263
지구 궤도 근처		
지구알베도로 인한 flux(W/m ²)	501	
지구 적외선으로 인한 flux(W/m ²)	237	
내부전자기기에서 발생하는 열에너지	100~300W	
S/C 의 전체면적	1m x 1.5m x 6	
radiation을 받는 부분	1m x 1.5m	
Finishes = black paint_Chemglaze Z306	$\alpha=0.92$	
	$\varepsilon=0.89$	

$$Q_{in} = Q_{out}$$

$$Q = \epsilon \sigma T^4$$

Q_{in} = Incident Direct Solar x Projected Area x Solar Absorptivity + α
 Q_{out} = Stephen-Boltzmann Constant x IR Emissivity x Total Area x (Temperature)⁴

최고 온도	9.38 °C
최저 온도	-94.2 °C

4. YAHOO 의 Thermal Control

- Thermal Sensor
- MLI
- Heat pipe & Radiator
- Patchy heater

Temperature Sensor



- Temperature range -60°C to +160°C

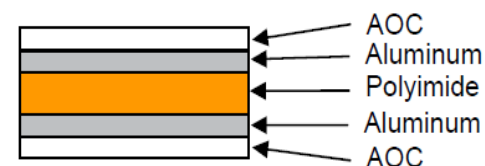
Multi Layer Insulation



A Multek Brand

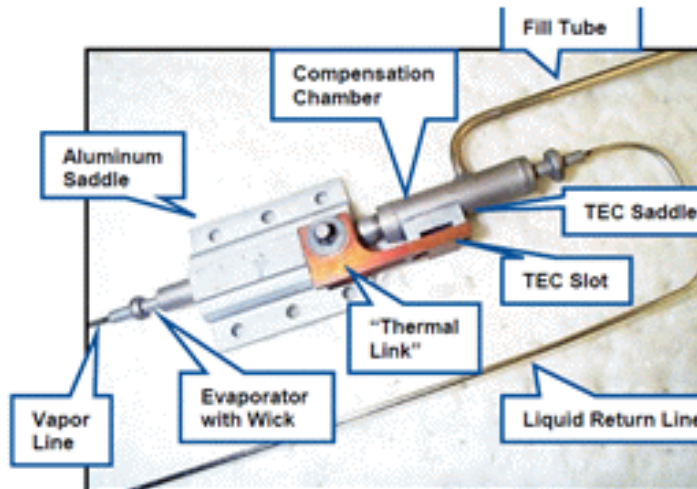
PRODUCT BULLETIN

**Aluminum Coated (Two Sides) Polyimide
with an Acrylic Overcoat**



Parameter (independent of film thickness)	Specified Value
Absorptance (α)	≤ 0.14
Hemispherical emittance (ϵ_H)	≤ 0.05
Normal emittance (ϵ_N)	≤ 0.05
Typical α/ϵ	4.0
Aluminum surface resistivity	$\leq 1 \Omega/\text{square}$ (before AOC application)
Intermittent temperature range	-250° C to 205° C (-420° F to 400° F)
Continuous temperature range	-250° C to 205° C (-420° F to 400° F)

Loop heat pipe & Radiator



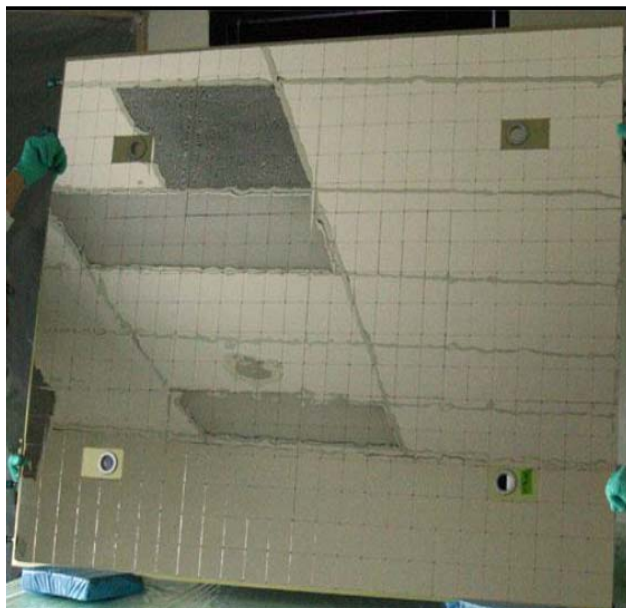
A Multek Brand

PRODUCT BULLETIN



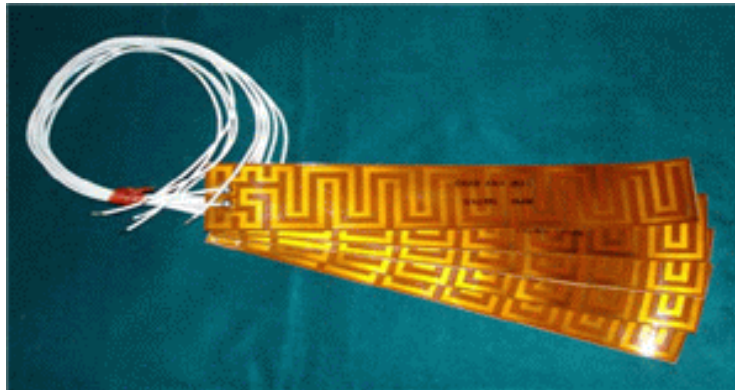
Silver Coated (One Side) FEP

Parameter (independent of film thickness)	Specified Value
Intermittent temperature range	-185° C to 260° C (-300° F to 500° F)
Continuous temperature range	-185° C to 150° C (-300° F to 300° F)



Standard Item Number	Thickness mil (μm)	Second Surface Mirror Properties			Typical Weight (g/m ²)	Item Number if Perforated
		α	εN	εH		
146399	0.5 (12.5)	≤ 0.09	≥ 0.40	≥ 0.40	28	TBD
146400	1.0 (25)	≤ 0.09	≥ 0.47	≥ 0.48	54	170769
146374	2.0 (51)	≤ 0.09	≥ 0.60	≥ 0.60	109	146736
146401	5.0 (127)	≤ 0.09	≥ 0.75	≥ 0.75	273	170203
146435	10.0 (254)	≤ 0.10	≥ 0.80	≥ 0.85	546	TBD

Heater



KAPTON®/SILICONE RUBBER FLEXIBLE HEATERS

Maximum Dimensions: 36" x 120"
Contact NPH for longer lengths Thickness: .006" - .055"
Termination area will be slightly larger
Temperature Range: -50°C to 230°C
Max Voltage: 600 Volts
Dielectric Strength: 1000VAC
Watt Density: 5 Watts per in/eq typical 60 Watts per in/eq Dependent on application
Wattage Tolerance: +5% - -10%

목차

Structure & Mechanism System

1. Structure & Mechanism System

2. 각 Subsystem의 Dimension, Mass

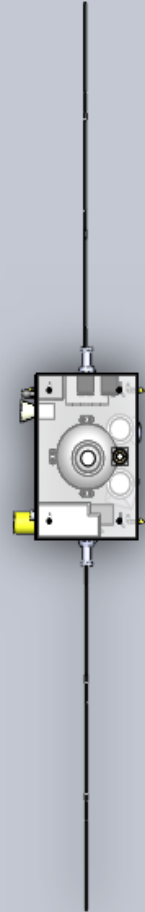
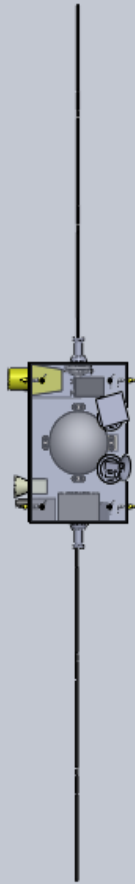
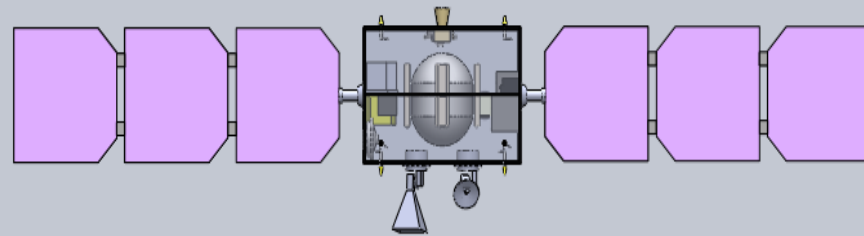
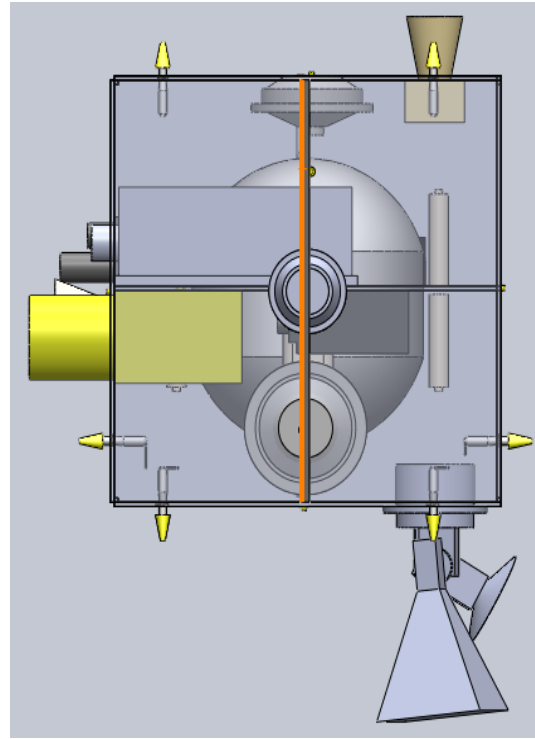
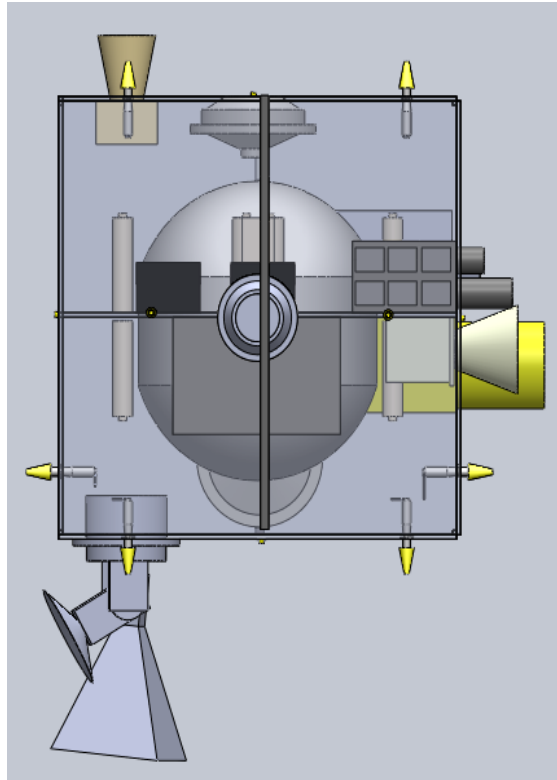
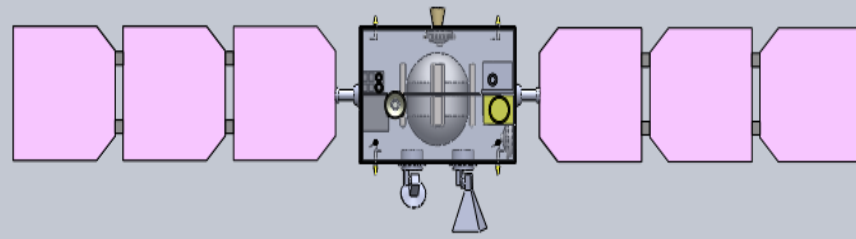
3. 각 Subsystem의 요구조건에 맞는 설계

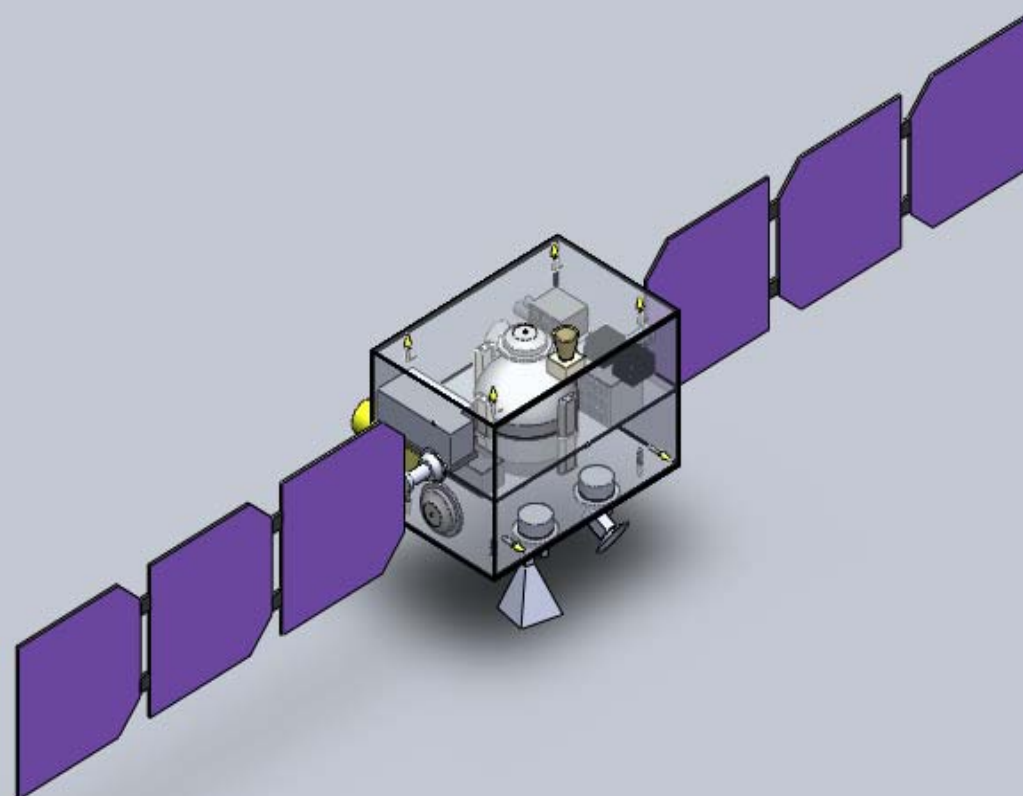
1. Structure & Mechanism System



1. 각 Subsystem의 Dimension, Mass

component		dimension(cm)	mass(kg)	수량
PAYLOAD	laser altimeter	22.2x16.5x26.5	10.24	1
	IR spectrometer	32x63x22	15.5	1
	thermal IR imager	16x31x14	3.3	1
	visual spectrometer	34x43x22	13.2	1
ACDS	sun sensor	2.3x0.9	0.014	8
	gyro	28.9x18x14.9	7.1	1
	star stracker	15.4x15.4x23.1	1.98	1
	thruster	8x17	0.3	12
	reaction wheel	31.6x15.9	6	4
TTC	antenna	27.4x30	3.3	1
	x band antenna	28x40	3.5	1
CNDH	Transponder	18x16x11	3.2	2
	computer	42x27x27	16	1
EPS	SADM	15.7x16	3	2
	battery	13x5x21	2.77	8
	solar panel	100x100	10	6
TCS	Heat pipe		5	1
	Radiator+Louver	30x30	8+8	1
	sensor		0.005	>20
	Heater		10	>20
SMS		150x100x100	100	
총합			261kg	





감사합니다
